Nonlocal Structured Sparsity Regularization Modeling for Hyperspectral Image Denoising

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***Abstract*— The nonlocal-based model for hyperspectral image (HSI) denoising first uses nonlocal self-similarity (NSS) prior to group similar full-band patches into 3-D nonlocal full-band groups (tensors) using a block matching (BM) operation, and then a low-rank (LR) penalty is typically applied to each nonlocal full-band group to reduce noise. While nonlocal-based methods have shown promising performance in HSI denoising, most existing methods have only considered the LR property of the nonlocal full-band group while ignoring the strong correlation between sparse coefficients. Moreover, such methods often result in unsatisfactory visual artifacts due to the noise sensitivity of BM operations, while requiring expensive computations. To address these limitations, this article proposes a novel nonlocal structured sparsity regularization (NLSSR) approach for HSI denoising. First, to mitigate the noise sensitivity of the BM operation, we propose a graph-based domain distance scheme to index similar full-band patches to form the nonlocal full-band group. Second, we design an adaptive unidirectional LR dictionary with low complexity that takes into account the differences in intrinsic structure correlation among different modes of the nonlocal full-band tensor. Third, we utilize a global spectral LR prior to reduce spectral redundancy. Fourth, we develop a generalized soft-thresholding (GST) algorithm based on the alternating minimization framework to solve the NLSSR-based HSI denoising problem. We perform extensive experiments on both simulated and real data to show that the proposed NLSSR algorithm outperforms many popular or state-of-the-art HSI denoising methods in both quantitative and visual evaluations.**

***Index Terms*— Alternating minimization algorithm, generalized soft-thresholding (GST), graph-based block matching (BM), hyperspectral image (HSI) denoising, low-rank (LR), nonlocal self-similarity (NSS), structured sparsity.**

I. Introduction

COMPARED to traditional grayscale or RGB images, hyperspectral imagery provides a more comprehensive acquisition of spectral data for the same scene, enabling precise differentiation of materials and facilitating its application in various fields, such as geological investigation [1], military monitoring [2], and terrain classification [3]. However, due to complicated uncertainty in the imaging process [4], hyperspectral image (HSI) is inevitably susceptible to noise, which not only degrades its visual quality, but also limits its usefulness in many subsequent tasks. Therefore, it is imperative to develop HSI denoising techniques to ensure reliable HSI analytics and understanding.

To date, numerous denoising methods have been proposed for HSI denoising [5], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16]. The most straightforward method involves applying effective gray-scale image denoising methods to restore each spectral channel of HSIs [17], [18]. However, such a method neglects the spectral correlation, which is a critical property of HSI data. More recent methods suggest reshaping the high-dimensional HSI data into a matrix and solving for its low-rank (LR) approximation to achieve denoising [8], [19]. Additionally, tensor decomposition methods have gained widespread use in HSI denoising, as an HSI can be represented as a third-order tensor [9], [10], [20], [21], [22], [23]. The two predominant tensor decomposition models, Tucker [24] and CANDECOMP/PARAFAC (CP) [25], have given rise to numerous HSI denoising algorithms. For instance, Renard et al. [9] proposed an LR tensor approximation (LRTA) model by employing Tucker decomposition for HSI denoising. Liu et al. [22] devised a PARAFAC model for HSI denoising by exploiting parallel factor analysis. Moreover, based on tensor singular value decomposition (t-SVD) [26], Zheng et al. [10] proposed a low-fibered-rank regularization model for HSI denoising. These LR and tensor decomposition approaches can adequately utilize global spectral information to reduce spectral redundancy. However, they often neglect critical spatial

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properties of HSIs, such as sparsity [27] and nonlocal self-similarity (NSS) [28], [29]. Consequently, these approaches usually produce suboptimal results characterized by spatial artifacts and noise.

The NSS prior highlights the recurring patterns of textures and structures in nonlocal regions of an image, and it can effectively preserve edges and sharpness [30]. Recent studies have integrated the NSS prior into HSI denoising [7], [27], [31], [32]. These approaches typically involve clustering similar full-band patches into 3-D nonlocal full-band groups (tensors) and then applying an LR penalty to each nonlocal full-band group. For example, Dong et al. [31] proposed a Laplacian scale mixture regularizer based on higher-order SVD for HSI denoising. Chen et al. [33] proposed a nonlocal tensor-ring (TR) approximation model for HSI denoising. Chang et al. [34] designed a weighted LR tensor recovery model, which has achieved superior performance for HSI denoising. Although these methods have achieved promising denoising performance, they usually suffer from some limitations, such as undesirable visual artifacts and severely time-consuming [35]. Recent advances have announced that integrating global spectral LR and NSS priors shows excellent potential in HSI denoising [7], [32], [35], [36]. A notable example is FastHyDe [35], which offers rapid processing but limited denoising results due to its lack of iterative refinement for the subspace and coefficient image. Recently, inspired by iterative regularization in image denoising [37], He et al. [7] developed a denoising paradigm that combines global spectral LR and spatial NSS priors. Zhuang et al. [32] proposed a global and nonlocal LR factorization approach for HSI denoising. While these nonlocal-based methods have shown impressive results in HSI denoising, they rely on the block matching (BM) operation [17], [38] to search for nonlocal similar patches, which are sensitive to noise and prone to generate undesirable visual artifacts in the denoised HSIs. For instance, when using NG-Meet [7], a state-of-the-art nonlocal-based HSI denoising approach, visual artifacts are evident in Fig. 1(e). Furthermore, most of these methods only consider the LR property of the nonlocal full-band group, ignoring the strong correlation between sparse coefficients.

Considering the aforementioned concerns, in this article, we propose a novel nonlocal structured sparsity regularization (NLSSR) approach, which jointly explores the global spectral and nonlocal structured sparsity priors for highly effective HSI denoising. Fig. 2 provides an overview of the proposed NLSSR approach for HSI denoising. The specific contributions of this article are as follows. First, a graph-based domain distance scheme is presented to compute patch similarity, mitigating the noise sensitivity of the BM operation. Second, we design an adaptive unidirectional LR dictionary with low complexity that fully considers the intrinsic structural correlations among different modes of the nonlocal full-band tensor. Third, we provide a generalized soft-thresholding (GST) algorithm based on the alternating minimization framework to optimize the proposed HSI denoising problem. Extensive experimental results are conducted on both simulated and real data to demonstrate that the proposed NLSSR denoising algorithm outperforms many popular or state-of-the-art HSI denoising methods in terms of both objective and perceptual quality.

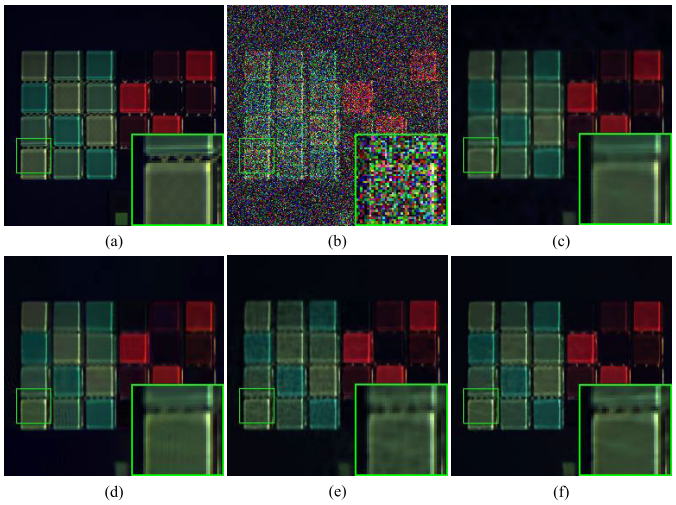


Fig. 1. Denoising results of MSI *Glass* on the CAVE dataset by different methods ($\sigma_n = 100$). (a) Original image (bands: 1, 8, and 15). (b) Noisy image. (c) LTDL [5]. (d) WLRTR [6]. (e) NG-Meet [7]. (f) Proposed NLSSR.

The remainder of this article is organized as follows. Section II describes some notations and preliminaries. Section III presents the proposed NLSSR model for HSI denoising. Section IV optimizes the proposed NLSSR-based HSI denoising problem. Section V presents the experimental results, and several concluding remarks are provided in Section VI.

II. Notations and Preliminaries

A. Notations

In this article, we denote the tensor, the matrix, the vector, and the scale as a calligraphic letter (e.g., $\mathcal{X}$), a boldface capital letter (e.g., $\boldsymbol{X}$), a bold lowercase letter (e.g., $\boldsymbol{x}$), and a lowercase letter (e.g., x), respectively. An N-dimensional tensor is denoted as $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_N}$, and each element $(i_1, i_2, \ldots, i_N)$ of the tensor $\mathcal{X}$ is defined as $x_{i_1 \times i_2 \times \cdots \times i_N}$. Each element (i, j) of the matrix $\boldsymbol{X}$ is defined as $x_{i,j}$, and the entry i of the vector $\boldsymbol{x}$ is defined as x_i. The Frobenius norm of the tensor $\mathcal{X}$ is defined as $\|\mathcal{X}\|_F = (\sum_{i_1=1}^{I_1} \sum_{i_2=1}^{I_2}, \ldots, \sum_{i_N=1}^{I_N} x_{i_1 \times i_2 \times \cdots \times i_N}^2)^{1/2}$. We denote the model-$n$ unfolding operation of the tensor $\mathcal{X}$ as $\boldsymbol{X}^{(n)} \in \mathbb{R}^{I_n \times (I_1, \ldots, I_{n-1} I_{n+1} I_N)}$, which is named as tensor matricization. Conversely, the unfolding matrix $\boldsymbol{X}$ along the n-mode can be folded as a tensor by $\mathcal{X} = \text{fold}_{(n)}(\boldsymbol{X})$. Moreover, the mode-$n$ multiplication of the tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_N}$ and the matrix $\boldsymbol{Y} \in \mathbb{R}^{I_n \times K}$ is denoted by $\mathcal{Z} = \mathcal{X} \times_n \boldsymbol{Y}$, where $\mathcal{Z} \in \mathbb{R}^{I_1 \times I_2, \ldots, K}$ and $\mathcal{X} \times_n \boldsymbol{Y} = \text{fold}_{(n)}(\boldsymbol{Y}\boldsymbol{X}^{(n)})$.

B. Preliminaries

In this section, we provide a brief overview of several concepts, such as the degradation model for HSI denoising, subspace LR representation model, BM operation, structural sparse coding (SSC) model, and graph-based transform.

1) Degradation Model for HSI Denoising: The goal of HSI denoising is to restore the original HSI $\mathcal{X} \in \mathbb{R}^{\mathcal{H} \times \mathcal{W} \times \mathcal{C}}$ (includes two spatial modes with the size of each band is

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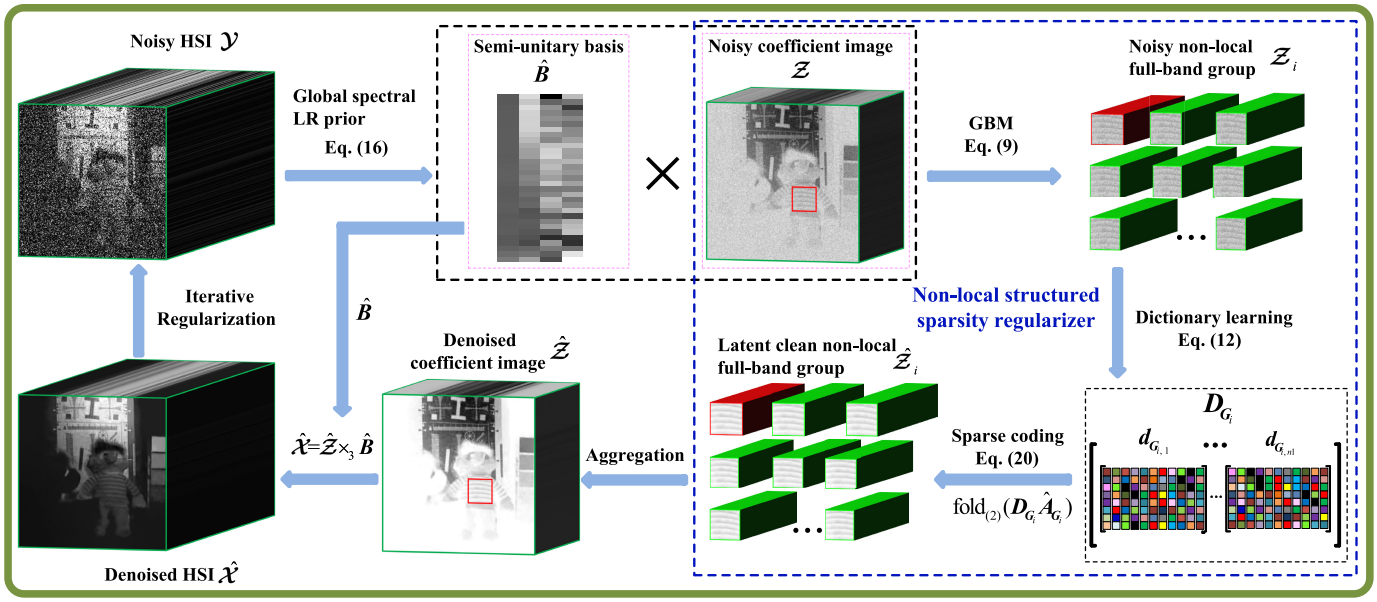


Fig. 2. Flowchart of the proposed NLSSR model for HSI denoising. The proposed NLSSR algorithm admits two complementary global spectral and nonlocal structured sparsity before taking advantage of the global LR and spatial NSS properties of HSIs.

$H \times W$ and one spectral mode with the number of bands is C) from its noisy observation $\mathcal{Y} \in \mathbb{R}^{\mathcal{H}\times\mathcal{W}\times\mathcal{C}}$, which is an ill-posed inverse problem and can be modeled as

$$\mathcal{Y} = \mathcal{X} + \mathcal{N} \tag{1}$$

where $\mathcal{N} \in \mathbb{R}^{\mathcal{H}\times\mathcal{W}\times\mathcal{C}}$ is usually considered to be zero-mean additive Gaussian noise [7], [34]. To address this ill-posed inverse problem, the most effective way is to exploit image prior knowledge of the original HSI $\mathcal{X}$ to regularize the solution space and thus solve the following minimization problem:

$$\hat{\mathcal{X}} = \arg\min_{\mathcal{X}} \frac{1}{2}\|\mathcal{Y} - \mathcal{X}\|_F^2 + \lambda\, \boldsymbol{R}(\mathcal{X}) \tag{2}$$

where $(1/2)\|\mathcal{Y} - \mathcal{X}\|_F^2$ represents the data fidelity term and λ is a balancing parameter. $\boldsymbol{R}(\mathcal{X})$ denotes the regularizer, which relies on the prior knowledge of the original scene [7], [8], [27]. Therefore, developing an effective regularizer to represent the HSI prior knowledge is crucial for effectively solving the HSI denoising problem.

2) Subspace LR Representation for HSIs: The HSI exhibits abundant spectral redundancy, indicating that images generated across various bands are often strongly correlated [39]. Therefore, the global spectral LR priors have been widely developed for HSI denoising [7], [32], [35], [36]. In HSI denoising, the subspace LR representation of a noise-free HSI $\mathcal{X} \in \mathbb{R}^{\mathcal{H}\times\mathcal{W}\times\mathcal{C}}$ can be expressed as

$$\mathcal{X} = \mathcal{Z} \times_3 \boldsymbol{B} \tag{3}$$

where $\boldsymbol{B} \in \mathbb{R}^{C\times M}$ ($M \ll C$) is a semiunitary basis matrix that captures the common subspace of different spectral bands and $\mathcal{Z} \in \mathbb{R}^{\mathcal{H}\times\mathcal{W}\times\mathcal{M}}$ denotes the coefficient image. The rationale of the subspace LR representation for an HSI is that the semiunitary basis matrix $\boldsymbol{B}$ is first learned by the HySime algorithm [40] or employing SVD of this HSI. Then, reconstructing the degraded HSI $\mathcal{Y}$ is translated into mining the prior knowledge of the coefficient image $\mathcal{Z}$. Consequently, the regularization model (2) for HSI denoising can be rewritten as

$$\{\hat{\mathcal{Z}}, \hat{\boldsymbol{B}}\} = \arg\min_{\mathcal{Z},\boldsymbol{B}} \frac{1}{2}\|\mathcal{Y} - \mathcal{Z} \times_3 \boldsymbol{B}\|_F^2 + \lambda\, \boldsymbol{R}(\mathcal{Z}) \quad \text{s.t. } \boldsymbol{B}^T\boldsymbol{B} = \boldsymbol{I} \tag{4}$$

where $\boldsymbol{I}$ denotes the identity matrix, and $\boldsymbol{R}(\mathcal{Z})$ is the regularizer, which is closely related to the coefficient image $\mathcal{Z}$. Once $\hat{\boldsymbol{B}}$ and $\hat{\mathcal{Z}}$ are obtained by solving (4), the latent clean image $\hat{\mathcal{X}}$ can be restored by $\hat{\mathcal{Z}} \times_3 \hat{\boldsymbol{B}}$.

3) BM Operation: The BM operation is an essential step in nonlocal-based image denoising algorithms [17], [38], which is a simple, yet effective approach to search for similar patches to a given exemplar one. Specifically, given an ordinary image $\boldsymbol{x} \in \mathbb{R}^N$, it is divided into n overlapped patches $\boldsymbol{x}_i \in \mathbb{R}^{\sqrt{b}\times\sqrt{b}}$, $i = 1, \ldots, n$. Taking each exemplar patch $\boldsymbol{x}_i$, the Euclidean distance is selected as the similarity criterion between different patches within a local window $L \times L$, that is,

$$d = \|\boldsymbol{x}_i - \boldsymbol{x}_k\|_2 \tag{5}$$

where $\boldsymbol{x}_k$ represents the kth candidate patch for the exemplar patch $\boldsymbol{x}_i$. We then choose m most similar patches, which are ranked in descending order of their Euclidean distance (i.e., the first m smallest values in all d values) to the exemplar patch $\boldsymbol{x}_i$, comprising the column of each data matrix $\boldsymbol{X}_i \in \mathbb{R}^{b\times m}$, that is, $\boldsymbol{X}_i = \{\boldsymbol{x}_{i,1}, \boldsymbol{x}_{i,2}, \ldots, \boldsymbol{x}_{i,m}\}$ with $\boldsymbol{x}_{i,m}$ being the mth similar vectorized patch.

4) SSC Model: For each exemplar patch $\boldsymbol{x}_i$ of the image $\boldsymbol{x}$ and based on the BM operation [17], [38], we can achieve a data matrix $\boldsymbol{X}_i$, which contains m patches with similar structures, is termed as a *nonlocal group*. Then, given a dictionary $\boldsymbol{D}_i$, each nonlocal group $\boldsymbol{X}_i$ is sparsely represented by solving the following minimization problem

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[41], [42], [43]:

$$\hat{\boldsymbol{A}}_i = \arg\min_{\boldsymbol{A}_i} \left(\frac{1}{2} \|\boldsymbol{X}_i - \boldsymbol{D}_i \boldsymbol{A}_i\|_F^2 + \tau \|\boldsymbol{A}_i\|_1 \right) \quad \forall i \tag{6}$$

where the notation $\|\cdot\|_F$ represents the Frobenius norm, $\|\cdot\|_1$ is utilized to sparsity penalty, and τ is a regularization parameter. The dictionary $\boldsymbol{D}_i$ is usually learned from the nonlocal group $\boldsymbol{X}_i$ [41]. Then, the whole image $\boldsymbol{x}$ can be sparsely represented by a set of group sparse codes $\{\hat{\boldsymbol{A}}_i\}_{i=1}^n$.

5) Graph-Based Transform: Defining a graph $\mathcal{G}(\mathcal{V}, \varrho, \boldsymbol{J})$, where $\mathcal{V}$ and $\boldsymbol{J}$ represent the sets of nodes and edges, respectively. The entry $w_{i,j}$ of the adjacency matrix $\boldsymbol{J}$ is a nonnegative value (except for 0 or 1), which is used to characterize the pairwise relationship between nodes i and j. Given a graph signal $\boldsymbol{x} \in \mathbb{R}^N$, which is one of the discrete samples of nodes $\mathcal{V}$ in a graph $\mathcal{G}$, where $N = |\mathcal{V}|$. Then the weight $w_{i,j}$ is assigned to each edge to feedback on the similarity between the pairwise adjacent nodes x_i and x_j, where $w_{i,j} = \exp(\|x_i - x_j\|_2^2 / \boldsymbol{\sigma}_w^2)$ and the parameter $\boldsymbol{\sigma}_w$ is used to control the noise sensitivity for the similarity measure between adjacent nodes [44]. We denote a degree matrix $\boldsymbol{G}$, which is a diagonal matrix. Each diagonal element $g_{i,i}$ is the degree of the node i, which is defined as $g_{i,i} = \sum_j w_{i,j}$. Then we can compute the graph Laplacian matrix as $\boldsymbol{L} = \boldsymbol{G} - \boldsymbol{J}$, and the graph-based transform $\boldsymbol{P}$ can be obtained by the eigenvectors of $\boldsymbol{L}$. Interested readers could refer to [45] for a detailed introduction to the graph-based transform.

III. HSI Denoising via the NLSSR Model

A. NLSSR Model for HSI Denoising

In this section, we propose a novel NLSSR model for HSI denoising, which jointly investigates the global spectral and nonlocal structured sparsity priors of the HSI. By integrating the proposed NLSSR regularizer into the global spectral LR framework (4), the NLSSR model for HSI denoising is represented as follows:

$$\{\hat{\mathcal{Z}}, \hat{\boldsymbol{B}}\} = \arg\min_{\mathcal{Z}, \boldsymbol{B}} \frac{1}{2} \|\mathcal{Y} \times_3 \boldsymbol{B}^T - \mathcal{Z}\|_F^2 + \lambda \|\mathcal{Z}\|_{\text{NLSSR}} \quad \text{s.t.}\ \boldsymbol{B}^T \boldsymbol{B} = \boldsymbol{I} \tag{7}$$

where $\|\mathcal{Z}\|_{\text{NLSSR}}$ represents the designed nonlocal-based structured sparsity regularizer, which will be introduced in detail below.

Given the degraded coefficient image $\mathcal{Z} \in \mathbb{R}^{\mathcal{H} \times \mathcal{W} \times \mathcal{M}}$ by decomposing the noisy HSI $\mathcal{Y} \in \mathbb{R}^{\mathcal{H} \times \mathcal{W} \times \mathcal{C}}$ using the global spectral LR prior [39], we first extract n overlapping full-band patches from $\mathcal{Z}$. Then, given each exemplar full-band patch $\mathcal{Z}_{\text{fbp}} \in \mathbb{R}^{\sqrt{b} \times \sqrt{b} \times M}$ of $\mathcal{Z}$ and by using the BM operation [17], [38], m similar nonlocal full-band patches within a local spatial window $L \times L$ are searched for constructing a nonlocal full-band group $\mathcal{Z}_i \in \mathbb{R}^{b \times m \times M}$, that is, $\mathcal{Z}_i = \{\boldsymbol{Z}_{i,1}^{(3)}, \boldsymbol{Z}_{i,2}^{(3)}, \ldots, \boldsymbol{Z}_{i,m}^{(3)}\}$, where $\boldsymbol{Z}_{i,m}^{(3)}$ is the mth similar nonlocal full-band patch of ith group $\mathcal{Z}_i$ and $\boldsymbol{Z}_{i,1}^{(3)} \in \mathbb{R}^{M \times b}$ is the unfolding version of $\mathcal{Z}_{\text{fbp}}$ in the spectral direction. Then, similar to the traditional SSC model (6) in Section II-B4, we can derive each nonlocal full-band group $\mathcal{Z}_i$ that can be sparsely represented by solving the following minimization problem:

$$\left\{\hat{\mathcal{G}}_i, \left\{\hat{\boldsymbol{D}}_i^{(k)}\right\}\right\} = \arg\min_{\hat{\mathcal{G}}_i, \{\hat{\boldsymbol{D}}_i^{(k)}\}} \frac{1}{2} \left\| \mathcal{Z}_i - \mathcal{G}_i \times_1 \boldsymbol{D}_i^{(1)} \times_2 \boldsymbol{D}_i^{(2)} \times_3 \boldsymbol{D}_i^{(3)} \right\|_F^2 + \tau \|\mathcal{G}_i\|_1 \quad \forall i \tag{8}$$

where $\mathcal{G}_i$ and $\boldsymbol{D}_i^{(k)}$ $(k = 1, 2, 3)$ represent the coefficient core tensor and the kth dimensional dictionary of the ith tensor group $\mathcal{Z}_i$, respectively. For each degraded full-band group $\mathcal{Z}_i$, once the coefficient core tensor $\hat{\mathcal{G}}_i$ and the dictionaries $\{\hat{\boldsymbol{D}}_i^{(k)}\}$ are obtained by solving (8), the latent clean coefficient image group $\hat{\mathcal{Z}}_i$ can be reconstructed as $\hat{\mathcal{Z}}_i = \hat{\mathcal{G}}_i \times_1 \hat{\boldsymbol{D}}_i^{(1)} \times_2 \hat{\boldsymbol{D}}_i^{(2)} \times_3 \hat{\boldsymbol{D}}_i^{(3)}$. After obtaining all coefficient image groups $\{\hat{\mathcal{Z}}_i\}_{i=1}^n$, we get the denoised coefficient image $\hat{\mathcal{Z}}$ by putting the groups back to their original locations and averaging the overlapped pixels. However, the above procedures for acquiring spatial information of the original image $\mathcal{X}$ usually suffer from some limitations: 1) since the BM operation is sensitive to noise, the denoised HSI is apt to produce undesirable visual artifacts and 2) the dictionary is learned from each mode successively, which inevitably leads to severe time consumption. We will address each of these limitations below.

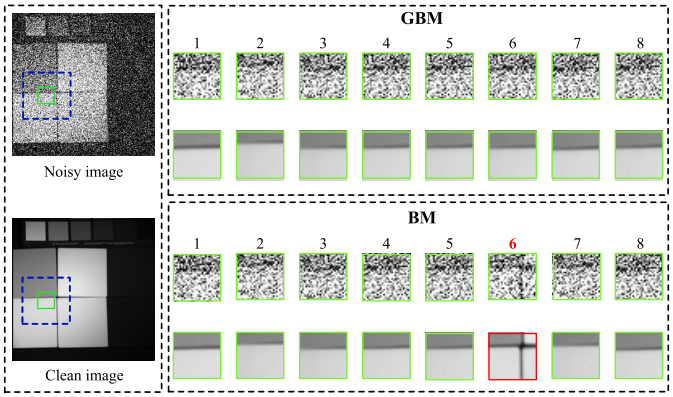


Fig. 3. Nonlocal similar patches selection using the proposed GBM and BM methods. For the same reference patch of the noisy image with $\boldsymbol{\sigma}_n = 100$, the proposed GBM obtains more accurate similar patch index results than the existing BM method.

B. Graph-Based BM

A critical step in nonlocal-based HSI denoising algorithms is the BM operation [17], [38], which typically uses Euclidean distance as a similarity criterion to identify nonlocal similar patches. However, Euclidean distance is highly sensitive to noise [44], [46], making it susceptible to producing undesirable visual artifacts in the denoised HSIs. In this section, we introduce a novel graph-based BM (GBM) strategy that employs graph-based domain distance for calculating patch similarity, instead of relying on the Euclidean distance (5) used in the traditional BM operation.

Similar to the BM operation for gray-scale images [17], [38], many existing HSI denoising methods also rely on spatial domain distance to gauge the similarity between different patches [7], [32], [34]. They first obtain a gray-scale

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image by averaging all bands of the given HSI. Then, they apply the BM operation to this average image and determine the spatial pixel positions of the most similar patches for each exemplar patch. This article also adopts this BM strategy for HSIs. As an alternative, we propose a graph-based domain distance approach for calculating the similarity between different patches, rather than using Euclidean distance (5). Specifically, given the degraded coefficient image $\mathcal{Z} \in \mathbb{R}^{\mathcal{H}\times\mathcal{W}\times\mathcal{M}}$, we first obtain an average image $\boldsymbol{Z} \in \mathbb{R}^{H\times W}$ through the spectral average operator. Then, this average image $\boldsymbol{Z}$ is divided into n overlapped patches $\boldsymbol{z}_i \in \mathbb{R}^{b\times 1}$, $i = 1, \ldots, n$. For each exemplar patch $\boldsymbol{z}_i$ within a local window $L \times L$, we have obtained m_c candidate patches $\{\boldsymbol{z}_{i,j}\}_{j=1}^{m_c}$, where $\boldsymbol{z}_{i,j} \in \mathbb{R}^{b\times 1}$ denotes the jth candidate patch of the ith exemplar patch $\boldsymbol{z}_i$. Then, by averaging out the exemplar patch $\boldsymbol{z}_i$ and its candidate patches $\{\boldsymbol{z}_{i,j}\}_{j=1}^{m_c}$, we obtain an average patch $\boldsymbol{z}_i^* \in \mathbb{R}^{b\times 1}$. Apparently, a similarity graph is constructed modeling the local neighborhood correlations among pixels in this average patch $\boldsymbol{z}_i^*$ [44]. We then apply the graph-based transform approach and a graph transform basis $\boldsymbol{P}_i \in \mathbb{R}^{b\times b}$ can be learned from the average patch $\boldsymbol{z}_i^*$ (for more details on learning the graph transform basis $\boldsymbol{P}_i$, see Section II-B5). To seek the targeted patches with similar structures for the exemplar patch $\boldsymbol{z}_i$, we transform the exemplar patch $\boldsymbol{z}_i$ and all candidate patches $\{\boldsymbol{z}_{i,j}\}_{j=1}^{m_c}$ into the graph domain through the graph transform basis $\boldsymbol{P}_i$. Then, we propose a graph-domain distance to measure the similarity between different patches as follows:

$$d = \|\boldsymbol{P}_i(\boldsymbol{z}_i - \boldsymbol{z}_{i,j})\|_1. \tag{9}$$

Through computing (9), the first m smallest distances in all d values are selected. Based on this, we can obtain the spatial pixels' position of the most m ($m < m_c$) nonlocal similar patches for each exemplar patch $\boldsymbol{z}_i$. Therefore, for each exemplar full-band patch $\mathcal{Z}_{\text{fbp}}$ in $\mathcal{Z}$ corresponding to $\boldsymbol{z}_i$ of the average image $\boldsymbol{Z}$, we can exploit the spatial pixels position of nonlocal similar patches $\{\boldsymbol{z}_{i,j}\}_{j=1}^{m}$ from the average image $\boldsymbol{Z}$ to construct a nonlocal full-band group $\mathcal{Z}_i \in \mathbb{R}^{b\times m\times M}$, that is, $\mathcal{Z}_i = \{\boldsymbol{Z}_{i,1}^{(3)}, \boldsymbol{Z}_{i,2}^{(3)}, \ldots, \boldsymbol{Z}_{i,m}^{(3)}\}$. The proposed GBM approach can mitigate the noise sensitivity issue as much as possible when searching for nonlocal similar patches. This improvement is attributed to the graph transform basis $\boldsymbol{P}_i$, which can effectively remove noise from the degraded patches, emphasizing textures and structures. This leads to more accurate similar patch index results compared to the traditional BM method. Fig. 3 illustrates the proposed GBM and BM methods for identifying nonlocal similar patches. It is evident that the clean patches obtained using the proposed GBM approach are highly similar, as determined by the spatial pixel positions of the corresponding noisy nonlocal similar patches. In contrast, clean patches found using the Euclidean distance (5) of the BM operation are inconsistent (e.g., the sixth patch). We will further validate the effectiveness of the proposed GBM method compared to the traditional BM approach in our experiments (refer to Section V-D for more details).

C. Adaptive Unidirectional LR Dictionary Learning

Let us revisit (8). It is apparent that designing an effective dictionary $\{\boldsymbol{D}_i^{(k)}\}$ is a critical step in the proposed nonlocal-based structured sparsity regularizer $\|\mathcal{Z}\|_{\text{NLSSR}}$. Over the past decade, many dictionary learning methods have been developed for HSI denoising, such as Tucker decomposition [9], CP decomposition [22], and block term decomposition [47]. These methods effectively preserve the spatial–spectral structures of HSIs. However, a common limitation is that they usually learn the dictionary for each mode successively, which can result in significant computational time requirements.

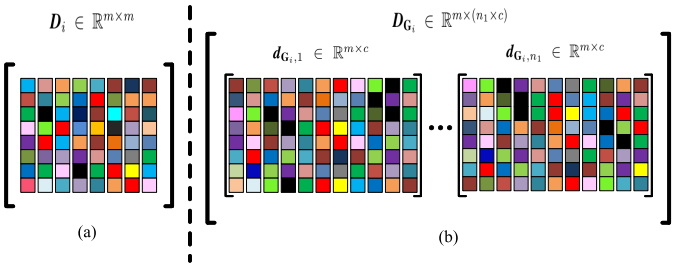


Fig. 4. Comparison between the dictionary learning in the (a) traditional SSC model [41], [42], [43] and the (b) proposed NLSSR model.

Recent advances have suggested that the nonlocal mode of the nonlocal full-band group $\mathcal{Z}_i$[1] has strong LR property, while the LR property of other modes is quite weak [34]. Toward this end, we devise a simple, yet effective adaptive unidirectional LR dictionary for the proposed nonlocal-based structured sparsity regularizer $\|\mathcal{Z}\|_{\text{NLSSR}}$. Specifically, the nonlocal full-band group $\mathcal{Z}_i \in \mathbb{R}^{b\times m\times M}$ is first unfolded into a matrix $\boldsymbol{Z}_i^{(2)} \in \mathbb{R}^{m\times c}$ at the nonlocal mode, where $c = b \times M$. Then, we perform SVD to $\boldsymbol{Z}_i^{(2)}$, namely

$$\boldsymbol{Z}_i^{(2)} = \boldsymbol{U}_{\mathbf{G}_i}\boldsymbol{\Sigma}_{\mathbf{G}_i}\boldsymbol{V}_{\mathbf{G}_i}^T = \sum_{j=1}^{n_1}\boldsymbol{\delta}_{\mathbf{G}_i,j}\boldsymbol{u}_{\mathbf{G}_i,j}\boldsymbol{v}_{\mathbf{G}_i,j}^T \tag{10}$$

where $\boldsymbol{\Sigma}_{\mathbf{G}_i} = \text{diag}(\boldsymbol{\delta}_{\mathbf{G}_i,1}, \boldsymbol{\delta}_{\mathbf{G}_i,2}, \ldots, \boldsymbol{\delta}_{\mathbf{G}_i,n_1})$ is a diagonal matrix and $n_1 = \min(m, c)$, and here $\boldsymbol{u}_{\mathbf{G}_i,j} \in \mathbb{R}^{m\times 1}$ and $\boldsymbol{v}_{\mathbf{G}_i,j} \in \mathbb{R}^{c\times 1}$ are the columns of $\boldsymbol{U}_{\mathbf{G}_i} \in \mathbb{R}^{m\times n_1}$ and $\boldsymbol{V}_{\mathbf{G}_i} \in \mathbb{R}^{c\times n_1}$, respectively.

After this, each dictionary atom $\boldsymbol{d}_{\mathbf{G}_i,j} \in \mathbb{R}^{m\times c}$ of the adaptive unidirectional LR dictionary $\boldsymbol{D}_{\mathbf{G}_i} \in \mathbb{R}^{m\times(n_1\times c)}$ for each nonlocal group $\boldsymbol{Z}_i^{(2)}$ is defined as

$$\boldsymbol{d}_{\mathbf{G}_i,j} = \boldsymbol{u}_{\mathbf{G}_i,j}\boldsymbol{v}_{\mathbf{G}_i,j}^T, \quad j = 1, 2, \ldots, n_1. \tag{11}$$

Then, we define the adaptive unidirectional LR dictionary $\boldsymbol{D}_{\mathbf{G}_i}$ for each nonlocal group $\boldsymbol{Z}_i^{(2)}$ as

$$\boldsymbol{D}_{\mathbf{G}_i} = [\boldsymbol{d}_{\mathbf{G}_i,1}, \boldsymbol{d}_{\mathbf{G}_i,2}, \ldots, \boldsymbol{d}_{\mathbf{G}_i,n_1}]. \tag{12}$$

In contrast to most existing multimode dictionary learning methods [9], [22], [47], [48], the designed adaptive unidirectional LR dictionary $\boldsymbol{D}_{\mathbf{G}_i}$ is a single mode-based dictionary learning approach with low complexity.[2] It fully considers the differences in intrinsic structure correlation among various modes of the nonlocal full-band tensor $\mathcal{Z}_i$. Fig. 4 shows the difference between dictionary learning in the traditional

[1]The nonlocal full-band group $\mathcal{Z}_i$ is a three-order tensor, which is usually formed by the spatial local sparsity domain in mode-1, nonlocal domain in mode-2, and spectral domain in mode-3.

[2]The computational complexity of the designed dictionary is $O(mc^2)$.

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SSC model [41], [42], [43] and the dictionary learning in the proposed NLSSR model. In general, in the traditional SSC model [41], [42], [43], the dictionary $\boldsymbol{D}_i$ is an orthogonal basis of size $m \times m$, which is directly learned from the group matrix $\boldsymbol{Z}_i^{(2)}$. This means that the atom size of the dictionary $\boldsymbol{D}_i$ is m. Compared with the dictionary $\boldsymbol{D}_i$ in the traditional SSC model [41], [42], [43], the size of each atom $\boldsymbol{d}_{\mathbf{G}_i,j}$ of the designed adaptive unidirectional LR dictionary $\boldsymbol{D}_{\mathbf{G}_i}$ is $m \times c$, which can better describe the spectral information of the nonlocal full-band group $\mathcal{Z}_i$. Then, based on the designed adaptive unidirectional LR dictionary learning strategy in (12), the proposed nonlocal-based structured sparsity regularizer $\|\mathcal{Z}\|_{\text{NLSSR}}$ in (7) can be represented as a simple, yet efficient SSC problem, that is,

$$\|\mathcal{Z}\|_{\text{NLSSR}} = \sum_{i=1}^{n} \frac{1}{2}\|\boldsymbol{Z}_i^{(2)} - \boldsymbol{D}_{\mathbf{G}_i}\boldsymbol{A}_{\mathbf{G}_i}\|_F^2 + \tau\|\boldsymbol{A}_{\mathbf{G}_i}\|_1. \quad (13)$$

Once the group sparse coefficient $\hat{\boldsymbol{A}}_{\mathbf{G}_i}$ is obtained by solving (13), the nonlocal group $\boldsymbol{Z}_i^{(2)}$ can be reconstructed as $\hat{\boldsymbol{Z}}_i^{(2)} = \boldsymbol{D}_{\mathbf{G}_i}\hat{\boldsymbol{A}}_{\mathbf{G}_i}$. Following this, the latent clean nonlocal full-band group $\hat{\mathcal{Z}}_i$ can be obtained by $\text{fold}_{(2)}(\hat{\boldsymbol{Z}}_i^{(2)})$. In (13), L_1-norm is used to sparsity penalty. However, it usually ignores the fact that for some real problems, including image inverse problems, it is difficult to achieve a satisfactory sparsity solution by solving the L_1 minimization problem [34], [49]. Therefore, we propose more effective sparse modeling by solving a weighted L_p minimization problem. Different from (13), the proposed regularizer model $\|\mathcal{Z}\|_{\text{NLSSR}}$ with the weighted L_p minimization is formulated as

$$\|\mathcal{Z}\|_{\text{NLSSR}} = \sum_{i=1}^{n} \frac{1}{2}\|\boldsymbol{Z}_i^{(2)} - \boldsymbol{D}_{\mathbf{G}_i}\boldsymbol{A}_{\mathbf{G}_i}\|_F^2 + \tau\|\boldsymbol{W}_{\mathbf{G}_i} \circ \boldsymbol{A}_{\mathbf{G}_i}\|_p \quad (14)$$

where $\circ$ represents the Hadamard product of two matrices, and $\|\cdot\|_p$ is the L_p-norm ($0< p \leq 1$). Note that we hereby enforce the p-norm for matrix, yet in effect, the L_p-norm is imposed on each column of $\boldsymbol{A}_{\mathbf{G}_i}$. $\boldsymbol{W}_{\mathbf{G}_i}$ is a weight matrix for each nonlocal group $\boldsymbol{Z}_i^{(2)}$, which can boost the representation ability of the group sparse coefficient $\hat{\boldsymbol{A}}_{\mathbf{G}_i}$. We describe how to solve the proposed NLSSR-based HSI denoising problem below.

IV. Optimization for the Proposed NLSSR Image Denoising Problem

Owing to the presence of multiple variables in the proposed denoising model (7), we adopt an alternating minimization strategy to make the optimization manageable. We address the objective function (7) by focusing on two subproblems, namely, the $\mathcal{Z}$ and $\boldsymbol{B}$ subproblems. Below, we present detailed implementations for obtaining effective solutions for each subproblem.

A. $\boldsymbol{B}$ Subproblem

Given the noisy HSI $\mathcal{Y}$, the $\boldsymbol{B}$ subproblem is reduced to

$$\hat{\boldsymbol{B}} = \underset{\boldsymbol{B}^T\boldsymbol{B}=\boldsymbol{I}}{\arg\min} \frac{1}{2}\|\mathcal{Y} - \mathcal{Z} \times_3 \boldsymbol{B}\|_F^2. \quad (15)$$

The rank-M SVD of $\mathcal{Y}^{(3)}$ is $\boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T$, then (15) has a closed-form solution [50]

$$\hat{\boldsymbol{B}} = \boldsymbol{V}, \quad \text{and } \bar{Z} = \text{fold}_{(3)}(\boldsymbol{U}\boldsymbol{\Sigma}). \quad (16)$$

For more details on solving the $\boldsymbol{B}$ subproblem, refer to [50].

B. $\mathcal{Z}$ Subproblem

After obtaining the solution of the $\boldsymbol{B}$ subproblem via (16), we can obtain an initialized reduced-dimensional coefficient image $\bar{Z} = \text{fold}_{(3)}(\boldsymbol{U}\boldsymbol{\Sigma})$. Through employing the proposed GBM strategy (see Section III-B), we can attain n nonlocal full-band groups $\{\bar{\mathcal{Z}}_i\}_{i=1}^n$. Then, each nonlocal full-band group $\bar{\mathcal{Z}}_i \in \mathbb{R}^{b\times m\times M}$ is unfolded into a matrix $\bar{\boldsymbol{Z}}_i^{(2)} \in \mathbb{R}^{m\times c}$ at the nonlocal mode. Through exploiting the designed adaptive unidirectional LR dictionary $\boldsymbol{D}_{\mathbf{G}_i}$ (see Section III-C) and invoking the proposed nonlocal-based structured sparsity regularizer $\|\mathcal{Z}\|_{\text{NLSSR}}$ in (14), then the $\mathcal{Z}$ subproblem is then translated into solving an SSC problem, namely

$$\hat{\boldsymbol{A}}_{\mathbf{G}_i} = \underset{\boldsymbol{A}_{\mathbf{G}_i}}{\arg\min}\left\{\frac{1}{2}\|\bar{\boldsymbol{Z}}_i^{(2)} - \boldsymbol{D}_{\mathbf{G}_i}\boldsymbol{A}_{\mathbf{G}_i}\|_F^2 + \tau\|\boldsymbol{W}_{\mathbf{G}_i} \circ \boldsymbol{A}_{\mathbf{G}_i}\|_p\right\} \quad \forall i. \quad (17)$$

To attain the solution of (17) effectively, we first introduce the following theorem.

Theorem 1:

$$\left\|\bar{\boldsymbol{Z}}_i^{(2)} - \bar{\mathbf{X}}_i^{(2)}\right\|_F^2 = \left\|\mathbf{R}_{\mathbf{G}_i} - \mathbf{A}_{\mathbf{G}_i}\right\|_F^2 \quad (18)$$

where $\bar{\mathbf{Z}}_i^{(2)} = \mathbf{D}_{\mathbf{G}_i}\mathbf{R}_{\mathbf{G}_i}$ and $\bar{\mathbf{X}}_i^{(2)} = \mathbf{D}_{\mathbf{G}_i}\mathbf{A}_{\mathbf{G}_i}$.

Proof: See the Appendix. □

Therefore, based on Theorem 1, the regularizer model (17) can be rewritten as

$$\hat{\boldsymbol{A}}_{\mathbf{G}_i} = \underset{\boldsymbol{A}_{\mathbf{G}_i}}{\arg\min}\left\{\frac{1}{2}\left\|\boldsymbol{R}_{\mathbf{G}_i} - \boldsymbol{A}_{\mathbf{G}_i}\right\|_F^2 + \tau\left\|\boldsymbol{W}_{\mathbf{G}_i} \circ \boldsymbol{A}_{\mathbf{G}_i}\right\|_p\right\} \quad \forall i. \quad (19)$$

It is evident that the optimization problem in (17) is transformed to address (19). To obtain an efficient solution for (19), we utilize the GST algorithm [51] for its resolution. To be concrete, given $\boldsymbol{R}_{\mathbf{G}_i}$, $\boldsymbol{W}_{\mathbf{G}_i}$ and p, a closed-form solution of (19) can be obtained as follows:

$$\hat{\boldsymbol{A}}_{\mathbf{G}_i} = \text{GST}(\boldsymbol{R}_{\mathbf{G}_i}, \tau\boldsymbol{W}_{\mathbf{G}_i}, p) \quad \forall i. \quad (20)$$

For more details about the GST algorithm, refer to [51]. Once the group sparse coefficient $\hat{\boldsymbol{A}}_{\mathbf{G}_i}$ is achieved by solving (20), the nonlocal group $\hat{\boldsymbol{Z}}_i^{(2)}$ can be reconstructed as $\hat{\boldsymbol{Z}}_i^{(2)} = \boldsymbol{D}_{\mathbf{G}_i}\hat{\boldsymbol{A}}_{\mathbf{G}_i}$. Following this, the latent clean nonlocal full-band group $\hat{\mathcal{Z}}_i$ can be attained by $\text{fold}_{(2)}(\hat{\boldsymbol{Z}}_i^{(2)})$. After achieving all nonlocal full-band groups $\{\hat{\mathcal{Z}}_i\}_{i=1}^n$, the denoised coefficient image $\hat{\mathcal{Z}}$ (i.e., the solution of $\mathcal{Z}$ subproblem) can be reconstructed by putting the groups $\{\hat{\mathcal{Z}}_i\}_{i=1}^n$ back to their original locations and averaging the overlapped pixels. Finally, we merge the denoised coefficient image $\hat{\mathcal{Z}}$ with $\hat{\boldsymbol{B}}$ to achieve the latent clean image $\hat{\mathcal{X}}$, namely, $\hat{\mathcal{X}} = \hat{\mathcal{Z}} \times_3 \hat{\boldsymbol{B}}$.

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C. Algorithm Refinement

In this section, we present some benefit strategies, including weight setting, regularization parameter setting, iterative regularization, and update low-dimensional subspace, to refine the proposed NLSSR algorithm to achieve better denoising performance.

1) Weight Setting: Given the fact that large values in the group sparse coefficient $\hat{\boldsymbol{A}}_{\mathbf{G}_i}$ usually represent major textures and edges. To effectively capture the spatial information of the original image, it is recommended to shrink larger values less while shrinking smaller values more [52]. Consequently, the weight $\boldsymbol{W}_{\mathbf{G}_i}$ of each group sparse coefficient $\hat{\boldsymbol{A}}_{\mathbf{G}_i}$ is set as follows:

$$\boldsymbol{W}_{\mathbf{G}_i}(k, j) = \frac{1}{|\boldsymbol{R}_{\mathbf{G}_i}(k, j)| + \varepsilon} \tag{21}$$

where $\boldsymbol{S}_{k,j}$ denotes the (k, j)th element of the arbitrary matrix $\boldsymbol{S}$, and ε is a small positive constant.

2) Regularization Parameter Setting: To make the proposed algorithm more stable, we adaptively determine the regularization parameter τ of (19) in each iteration to balance the fidelity term and the regularization term. Specifically, inspired by [53], and τ is set as follows:

$$\tau = \frac{2\sqrt{2}\ \boldsymbol{\sigma}_n^{2^{(t)}}}{(\boldsymbol{\varphi}_i + \epsilon)} \tag{22}$$

where t represents the tth iteration. φ_i represents the estimated standard variance of $\boldsymbol{R}_{\mathbf{G}_i}$ and ϵ is a small positive constant.

3) Iterative Regularization: Iterative regularization [37] is a powerful technique for improving image denoising performance and has been widely adopted to enhance the performance of denoising algorithms [18], [41], [43]. In this article, we also employ iterative regularization to update the subsequent input noisy HSI and reestimate the noise variance $\boldsymbol{\sigma}_n^2$, thereby refining the semiunitary basis $\boldsymbol{B}$ in the subspace LR learning stage. The fundamental concept of iterative regularization is to incorporate the filtered noise into the denoised image. As a result, the next input noisy HSI is updated as follows:

$$\mathcal{Y}^{(t+1)} = \mu\hat{\mathcal{X}}^{(t)} + (1 - \mu)\mathcal{Y} \tag{23}$$

where μ is a relaxation parameter, which is used to tradeoff the denoised HSI $\hat{\mathcal{X}}$ and the original noisy HSI $\mathcal{Y}$.

As the noise level is changed during the iteration, the noise standard deviation $\boldsymbol{\sigma}_n$ is required to recompute when exploiting the proposed nonlocal-based structured sparsity regularizer $\|\mathcal{Z}\|_{\text{NLSSR}}$ (14) to solve the $\mathcal{Z}$ subproblem. Since the subspace basis $\boldsymbol{B}$ of (15) is semiunitary, we can claim that the coefficient image $\mathcal{Z}$ has the same noise level as the noisy image $\mathcal{Y}$ [7], [35]. Therefore, we extend the idea of iterative regularization to update the estimation of noise standard deviation $\boldsymbol{\sigma}_n$. Specifically, the standard deviation of noise $\boldsymbol{\sigma}_n$ in the tth iteration is formulated as [37]

$$\boldsymbol{\sigma}_n{}^{(t)} = \rho\sqrt{\left(\boldsymbol{\sigma}_n^2 - \left\|\mathcal{Y} - \mathcal{Y}^{(t+1)}\right\|_F^2\right)} \tag{24}$$

where ρ is a scaling factor used to control the reestimation of noise variance. In particular, under the severe contamination of the input image, the subspace basis $\boldsymbol{B}$ can be effectively refined by this iterative regularization strategy.

Algorithm 1 Proposed NLSSR for HSI Denoising

Require: Noisy image $\mathcal{Y}$.

Initialize $\hat{\mathcal{X}}^0 = \mathcal{Y}$, $\mathcal{Y}^0 = \mathcal{Y}$, $\boldsymbol{\sigma}_n$ and estimate the rank M by HySime [40].
for $t = 1$ **to** Max-Iter **do**
  Learn the subspace basis $\hat{\boldsymbol{B}}$ by computing (16).
  Estimate the reduced-dimensional coefficient image $\bar{Z}$ by $\bar{Z} = \text{fold}_{(3)}(\boldsymbol{U}\boldsymbol{\Sigma})$.
  for Each full-band patch $\bar{Z}_{fbp}$ in $\bar{Z}$ **do**
    Find nonlocal similar full-band patches to form a nonlocal full-band group $\bar{Z}_i$ using (9).
    The nonlocal full-band group $\bar{Z}_i$ is unfolded into a matrix $\bar{\boldsymbol{Z}}_i^{(2)}$ at the nonlocal mode.
    Learn the dictionary $\boldsymbol{D}_{\mathbf{G}_i}$ from $\bar{\boldsymbol{Z}}_i^{(2)}$ using (12).
    Update the regularization parameter τ by computing (22).
    Update the weight $\boldsymbol{W}_{\mathbf{G}_i}$ by computing (21).
    Estimate $\hat{\boldsymbol{A}}_{\mathbf{G}_i}$ by computing (20).
    Get the estimation: $\hat{\boldsymbol{Z}}_{\mathbf{G}_i} = \boldsymbol{D}_{\mathbf{G}_i}\hat{\boldsymbol{A}}_{\mathbf{G}_i}$.
    The latent clean nonlocal full-band group $\hat{\mathcal{Z}}_i$ can be attained by $\text{fold}_{(2)}(\hat{\boldsymbol{Z}}_i^{(2)})$.
  end for
  Aggregate all $\{\hat{\mathcal{Z}}_i\}_{i=1}^n$ to form the denoised coefficient image $\hat{\mathcal{Z}}$.
  The latent clean image $\hat{\mathcal{X}}$ can be reconstructed by $\hat{\mathcal{X}} = \hat{\mathcal{Z}} \times_3 \hat{\boldsymbol{B}}$.
  Update iterative regularization by computing (23).
  Re-estimate the noise standard deviation $\boldsymbol{\sigma}_n$ by computing (24).
  Update the rank M of subspace $\boldsymbol{B}$ by computing (25).
end for
Output: The final denoised HSI $\hat{\mathcal{X}}$.

4) Update Low-Dimensional Subspace: Since the noise level is changed in each iteration, we also need to update the rank M of the subspace basis $\boldsymbol{B}$ with the iteration, which plays a tradeoff role in preserving image information (higher rank values) and removing noise (lower rank values). In general, the input degraded HSI with severe noise, the rank M should be set to a small value [50], which can effectively eliminate the noise and lead to a promising denoising result. Therefore, we first apply HySime [40] to obtain an initialized small value M. With the increase of iteration numbers, the noise intensity is gradually descending. In other words, the larger the noise intensities, the smaller the rank M more. Besides, the rank M never exceeds the band numbers of the original HSI C. Therefore, we are heuristically update the rank M of subspace basis $\boldsymbol{B}$ as follows:

$$M^{(t+1)} = \min(M^{(t)} + \gamma \times t, C) \tag{25}$$

where γ is a constant, which is used as a tradeoff between the iteration numbers and the rank M.

Thus far, we have provided a detailed description of the alternating minimization strategy to address the proposed

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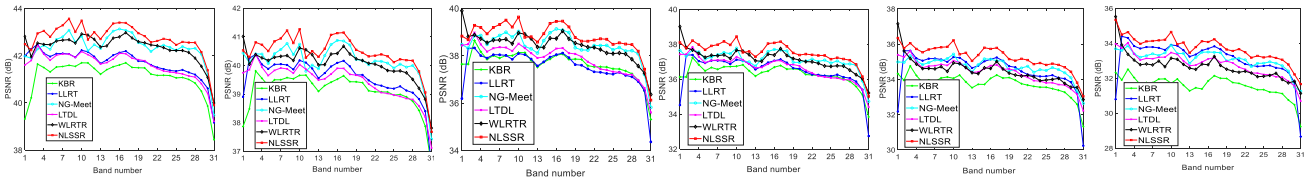


Fig. 5. Average PSNR values of each band on the CAVE dataset at different noise levels. (a)–(f) $\sigma_n = 20$, 30, 40, 50, 75, and 100, respectively.

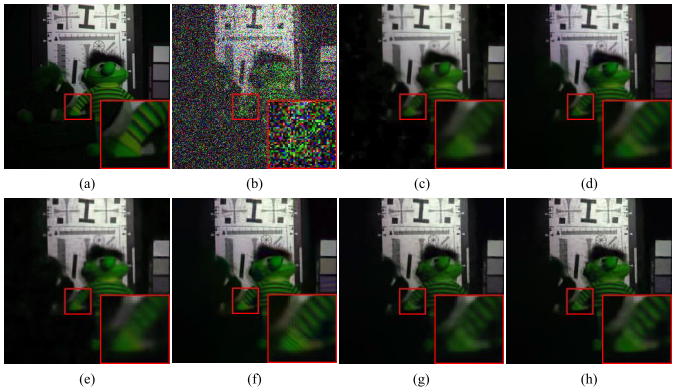


Fig. 6. Denoising results of MSI *Toy* on the CAVE dataset by different methods ($\sigma_n = 100$). (a) Original image (bands: 6, 14, and 30). (b) Noisy image. (c) KBR [27]. (d) LLRT [34]. (e) LTDL [5]. (f) WLRTR [6]. (g) NG-Meet [7]. (h) **NLSSR**.

NLSSR-based HSI denoising problem. By solving each subproblem independently, we can achieve a stable solution. Additionally, we have presented four advantageous strategies to enhance the proposed algorithm, making it more effective and efficient. We summarize the complete algorithm of the proposed NLSSR model for HSI denoising in Algorithm 1.

V. Experimental Results

In this section, extensive experiments are conducted to verify the effectiveness of the proposed NLSSR model for HSI denoising. We apply the proposed NLSSR denoising algorithm to simulated and real HSIs, respectively. The HSI consists of two parts, that is, spatial and spectral, which both play a vital role in denoised HSIs. In this article, we employ five quantitative image quality metrics to assess the quality of denoised HSIs, including the peak signal-to-noise ratio (PSNR), structural similarity (SSIM) [54], feature similarity (FSIM) [55], erreur relative globale adimensionnelle de synthese (ERGAS) [56], and spectral angle mapper (SAM) [57]. PSNR and SSIM are two prevalent spatial-based quality indices for image restoration that measure the similarity between the target image and the reference image in terms of mean squared error (MSE) and structural consistency, respectively. FSIM elaborates on the perceptual consistency between the two images being compared. Higher values for PSNR, SSIM, and FSIM indicate better-denoised image quality. ERGAS is a well-established spectral-based quality index, measuring the fidelity of the denoised HSI based on the weighted sum of MSE in each band. SAM computes the

TABLE I

Average Denoising Results of PSNR (dB), SSIM, FSIM, ERGAS, and SAM by KBR [27], LLRT [34], LTDL [5], WLRTR [6], NG-Meet [7], and the Proposed NLSSR Methods on the CAVE Dataset

σ_n	Index	KBR	LLRT	LTDL	WLRTR	NG-Meet	**NLSSR**
20	PSNR↑	41.11	41.69	41.60	42.35	42.35	**42.73**
	SSIM↑	0.9639	0.9743	0.9749	0.9754	0.9786	**0.9793**
	FSIM↑	0.9721	0.9688	0.9739	0.9694	**0.9770**	0.9767
	ERGAS↓	50.34	46.94	47.23	43.81	44.15	**41.88**
	SAM↓	0.1880	0.1036	0.1016	0.1042	0.0963	**0.0921**
30	PSNR↑	39.09	39.55	39.42	40.04	40.13	**40.43**
	SSIM↑	0.9469	0.9622	0.9622	0.9614	0.9664	**0.9674**
	FSIM↑	0.9624	0.9587	0.9636	0.9590	**0.9687**	0.9686
	ERGAS↓	63.46	60.21	60.61	57.01	56.58	**54.44**
	SAM↓	0.2231	0.1252	0.1232	0.1302	0.1169	**0.1139**
40	PSNR↑	37.65	37.52	37.87	38.43	38.43	**38.78**
	SSIM↑	0.9384	0.9425	0.9495	0.9467	0.9537	**0.9556**
	FSIM↑	0.9516	0.9465	0.9541	0.9507	0.9616	**0.9617**
	ERGAS↓	75.54	75.83	72.47	68.45	68.92	**65.82**
	SAM↓	0.2023	0.1566	0.1441	0.1551	0.1461	**0.1386**
50	PSNR↑	36.30	36.48	36.64	37.14	37.12	**37.53**
	SSIM↑	0.9125	0.9320	0.9366	0.9313	0.9403	**0.9429**
	FSIM↑	0.9424	0.9389	0.9450	0.9436	0.9538	**0.9539**
	ERGAS↓	88.05	85.82	83.49	79.22	79.99	**75.90**
	SAM↓	0.2699	0.1698	0.1634	0.1786	0.1655	**0.1552**
75	PSNR↑	33.74	34.52	34.40	34.48	34.71	**35.28**
	SSIM↑	0.8719	0.9062	0.9057	0.8842	0.9103	**0.9168**
	FSIM↑	0.9190	0.9217	0.9247	0.9229	0.9377	**0.9389**
	ERGAS↓	117.92	108.02	108.10	107.56	104.98	**97.87**
	SAM↓	0.3151	0.2198	0.2173	0.2421	0.2229	**0.1982**
100	PSNR↑	31.56	33.09	32.79	32.67	33.10	**33.73**
	SSIM↑	0.8296	0.8820	0.8755	0.8379	0.8820	**0.8909**
	FSIM↑	0.8937	0.9070	0.9073	0.9048	0.9218	**0.9244**
	ERGAS↓	151.24	127.20	130.26	131.93	126.04	**116.96**
	SAM↓	0.3465	0.2222	0.2554	0.2927	0.2606	**0.2305**
Average	PSNR↑	36.57	37.14	37.12	37.52	37.64	**38.08**
	SSIM↑	0.9105	0.9332	0.9341	0.9228	0.9385	**0.9421**
	FSIM↑	0.9402	0.9402	0.9448	0.9417	0.9534	**0.9540**
	ERGAS↓	91.09	84.00	83.70	81.33	80.11	**75.48**
	SAM↓	0.2575	0.1662	0.1675	0.1838	0.1681	**0.1548**

average spectral difference between the target HSI and the reference HSI. Unlike the aforementioned spatial-based quality indices, lower ERGAS and SAM values signify superior denoising performance.

Before running the proposed NLSSR-based HSI denoising algorithm (see Algorithm 1), we normalize all HSIs to [0255]. In the proposed NLSSR algorithm, there are several hyperparameters that should be adjusted, including patch size b, nonlocal similar patches m, the power value p, iterative regularization parameters ρ and μ, the spectral dimension M, and the subspace update parameter γ. We set the patch size $b = 5$, as well as nonlocal similar patches m and the power p are set to (150, 0.95) and (200, 0.90) for $\sigma_n \leq 20$ and $\sigma_n > 20$, respectively. We empirically set $\rho = 0.5$ and $\mu = 0.1$ in all experiments. The spectral dimension M is an important parameter, which is initialized by HySime [40] and updated by (25). The subspace update parameter γ is fixed to 2 in the whole experiments, as introduced in [50]. The source code of the proposed NLSSR model for HSI denoising is available at https://github.com/zhazhiyuan/NLSSR_Denoising_Demo.git.

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TABLE II

DENOISING RESULTS OF PSNR (dB), SSIM, FSIM, ERGAS, AND SAM BY LRTA [9], BM4D [29], KBR [27], FASTHYDE [35], NLR-CPTD [58], SLRL4D [36], LTDL [5], WLRTR [6], GLF [32], LRTFD-FR [59], NG-MEET [7], AND NLSSR METHODS ON THE PAC DATASET

σ_n	Index	LRTA	BM4D	KBR	FastHyDe	NLR-CPTD	SLRL4D	LTDL	WLRTR	GLF	LRTFD-FR	NG-Meet	**NLSSR**
20	PSNR↑	34.16	35.01	36.83	38.91	35.10	37.11	39.18	38.84	39.38	38.52	39.61	**39.96**
	SSIM↑	0.9134	0.9397	0.9645	0.9752	0.9492	0.9692	0.9777	0.9765	0.9781	0.9675	0.9798	**0.9812**
	FSIM↑	0.9600	0.9630	0.9779	0.9852	0.9681	0.9819	0.9869	0.9852	0.9864	0.9840	0.9876	**0.9883**
	ERGAS↓	84.67	76.59	61.91	49.67	75.60	56.33	47.75	49.66	46.98	51.43	44.99	**43.23**
	SAM↓	0.1419	0.1115	0.0706	0.0766	0.0849	0.0758	0.0646	0.0658	0.0681	0.0800	0.0591	**0.0588**
30	PSNR↑	31.79	32.43	34.44	36.60	34.13	35.17	36.72	36.41	36.90	34.91	37.00	**37.41**
	SSIM↑	0.8683	0.8934	0.9408	0.9604	0.9352	0.9532	0.9634	0.9613	0.9639	0.9276	0.9648	**0.9681**
	FSIM↑	0.9337	0.9340	0.9620	0.9758	0.9600	0.9713	0.9780	0.9749	0.9770	0.9673	0.9782	**0.9798**
	ERGAS↓	110.38	102.85	81.36	64.08	84.41	70.21	63.06	65.31	61.92	77.45	60.60	**57.82**
	SAM↓	0.1631	0.1348	0.0847	0.0916	0.1004	0.0870	0.0804	0.0806	0.0846	0.1297	0.0774	**0.0769**
40	PSNR↑	30.12	30.69	31.65	34.99	33.02	33.59	34.92	34.77	35.16	30.84	35.09	**35.75**
	SSIM↑	0.8208	0.8445	0.8886	0.9449	0.9161	0.9349	0.9473	0.9453	0.9483	0.8331	0.9474	**0.9547**
	FSIM↑	0.9072	0.9032	0.9283	0.9659	0.9487	0.9586	0.9682	0.9644	0.9671	0.9294	0.9677	**0.9718**
	ERGAS↓	133.81	125.46	112.31	76.70	95.94	84.20	77.21	78.57	75.22	124.67	75.41	**69.93**
	SAM↓	0.1806	0.1534	0.1046	0.1041	0.1176	0.0974	0.0906	0.0933	0.0979	0.2196	0.0900	**0.0893**
50	PSNR↑	28.79	29.42	30.63	33.78	31.88	32.52	33.53	33.51	33.84	27.90	33.69	**34.37**
	SSIM↑	0.7706	0.7963	0.8605	0.9298	0.8924	0.9175	0.9287	0.9284	0.9317	0.7265	0.9287	**0.9391**
	FSIM↑	0.8811	0.8732	0.9090	0.9561	0.9350	0.9474	0.9576	0.9537	0.9566	0.8867	0.9563	**0.9619**
	ERGAS↓	156.14	145.07	126.37	87.91	109.33	95.05	90.45	90.73	87.39	176.65	88.62	**81.91**
	SAM↓	0.1981	0.1692	0.1157	0.1147	0.1345	0.1049	0.1095	0.1048	0.1090	0.3032	0.1047	**0.1008**
75	PSNR↑	26.58	27.35	28.44	31.66	29.46	30.55	31.04	31.00	31.46	27.44	31.45	**32.06**
	SSIM↑	0.6559	0.6886	0.7733	0.8920	0.8208	0.8716	0.8772	0.8784	0.8871	0.7092	0.8859	**0.9024**
	FSIM↑	0.8249	0.8073	0.8483	0.9322	0.8963	0.9184	0.9273	0.9233	0.9290	0.8775	0.9298	**0.9388**
	ERGAS↓	202.30	184.06	163.08	112.06	144.41	119.25	120.33	120.95	114.66	183.61	114.75	**106.89**
	SAM↓	0.2277	0.2005	0.1411	0.1370	0.1757	0.1259	0.1276	0.1357	0.1317	0.2683	0.1260	**0.1244**
100	PSNR↑	25.06	26.10	27.38	30.27	27.81	29.22	29.13	29.40	29.89	25.97	29.84	**30.56**
	SSIM↑	0.5469	0.6032	0.7170	0.8540	0.7460	0.8270	0.8082	0.8287	0.8396	0.6374	0.8368	**0.8663**
	FSIM↑	0.7739	0.7571	0.8086	0.9098	0.8550	0.8908	0.8844	0.8950	0.9012	0.8448	0.8998	**0.9180**
	ERGAS↓	241.93	212.76	184.50	131.52	174.56	138.94	150.44	145.44	137.52	216.41	138.13	**127.02**
	SAM↓	0.2422	0.2247	0.1602	0.1501	0.1996	0.1465	0.1449	0.1603	0.1491	0.2893	0.1458	**0.1407**
Average	PSNR↑	29.42	30.17	31.56	34.37	31.90	33.03	34.09	33.99	34.44	30.93	34.45	**35.02**
	SSIM↑	0.7627	0.7943	0.8575	0.9261	0.8766	0.9122	0.9171	0.9198	0.9248	0.8002	0.9239	**0.9353**
	FSIM↑	0.8801	0.8730	0.9057	0.9542	0.9272	0.9447	0.9504	0.9494	0.9529	0.9150	0.9532	**0.9598**
	ERGAS↓	154.87	141.13	121.59	86.99	114.04	94.00	91.54	91.78	87.28	138.37	87.08	**81.13**
	SAM↓	0.1923	0.1657	0.1128	0.1123	0.1355	0.1063	0.1029	0.1068	0.1067	0.2150	0.1005	**0.0985**

A. Simulated Experiments

In the simulated experiments, we apply three popular datasets to evaluate the denoising performance of the proposed NLSSR algorithm, including the Columbia multispectral image dataset (CAVE),[3] the Pavia City Center hyperspectral dataset (PAC),[4] and the Washington DC Mall dataset (WDC).[5] We add zero-mean additive white Gaussian noise to images of these datasets to generate noisy observations. Due to the limited space, we only present the denoising results at six noise levels, that is, Gaussian white noise with standard deviation $\sigma_n = \{20, 30, 40, 50, 75, 100\}$.

1) Simulated Experiments for the CAVE Dataset: The CAVE dataset possesses 32 real-world objects and materials. Each multispectral image (MSI) in the CAVE dataset is composed of spatial size 512×512 and spectral dimension 31. Since several compared methods are running very slowly, we shrink all MSIs of the CAVE dataset to subimages with size $256 \times 256 \times 31$. To ensure a fair comparison of the CAVE dataset, we evaluate our proposed NLSSR method against five state-of-the-art MSI denoising methods, namely KBR [27], LLRT [34], LTDL [5], WLRTR [6], and NG-Meet [7], as these methods have been previously tested on the CAVE dataset in their respective studies. It is worth mentioning that KBR, LLRT, LTDL, and WLRTR are spatial nonlocal-based methods, while the NG-Meet technique integrates global spectral LR and spatial NSS priors. We use the original code implementation of these competing methods and employ either their default parameter setting or the suggested parameters in their papers to obtain the denoising results. We focus on the same denoising task and do not exhaustively optimize their results, as their parameters are typically the best settings. Through generating noisy observations, then all competing methods are performed to restore the latent clean HSI. The average quantitative results of the CAVE dataset are presented in Table I, with the best results highlighted in bold. An overall impression observed from Table I is that the proposed NLSSR outperforms other compared methods based on all quantitative quality metrics (the only exception is when $\sigma_n = 20$ and 30 for which NG-Meet is slightly higher than the proposed method in terms of FSIM). In particular, the average PSNR gains of the proposed NLSSR over KBR, LLRT, LTDL, WLRTR, and NG-Meet methods are 1.51, 0.94, 0.96, 0.56, and 0.44, respectively. Meanwhile, the proposed NLSSR enjoys an ERGAS reduction over KBR by 15.61, over LLRT by 8.53, over LTDL by 8.22, over WLRTR by 5.85, and over NG-Meet by 4.63. Furthermore, Fig. 5 displays the average PSNR values for each band on the CAVE dataset at six different noise levels. The results show that our proposed NLSSR method (red line) outperforms the other compared methods, as it achieves higher PSNR values across all bands.

The visual comparison results of MSI *Toy* with noise standard deviation $\sigma_n = 100$ are shown in Fig. 6. To better demonstrate the denoising results of all competing methods, we have magnified the subregion of each denoised image to compare them. We can see that LLRT and WLRTR methods

[3]https://www.cs.columbia.edu/CAVE/databases/multispectral/

[4]http://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes

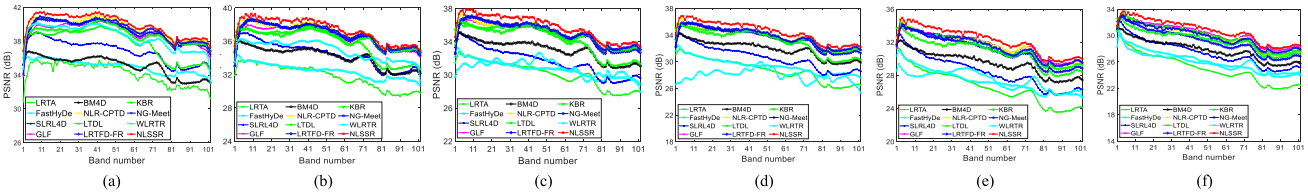
[5]https://engineering.purdue.edu/~biehl/MultiSpec/hyperspectral.html

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TABLE III

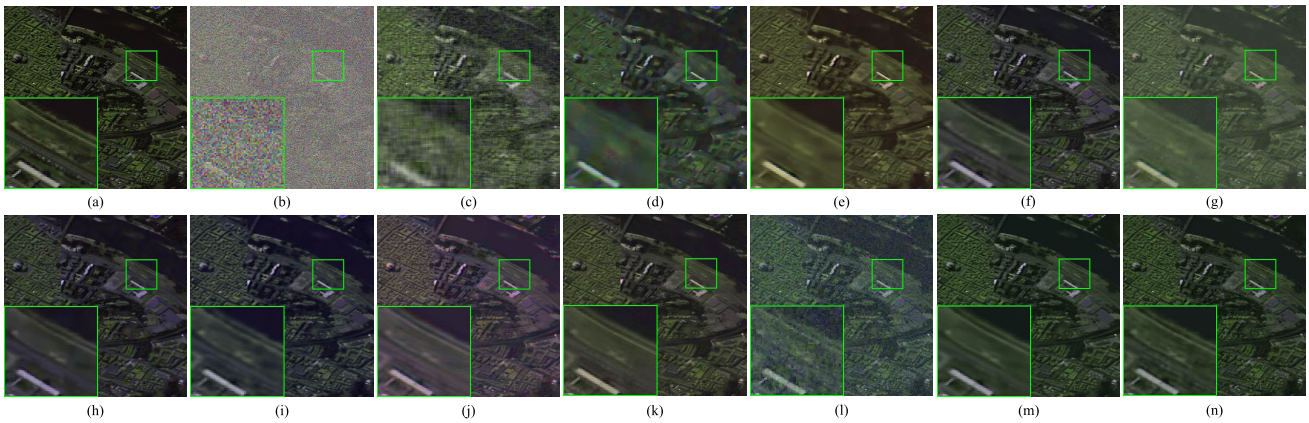
DENOISING RESULTS OF PSNR (dB), SSIM, FSIM, ERGAS, AND SAM BY LRTA [9], BM4D [29], KBR [27], FASTHYDE [35], NLR-CPTD [58], SLRL4D [36], LTDL [5], WLRTR [6], GLF [32], LRTFD-FR [59], NG-MEET [7], AND NLSSR METHODS ON THE WDC DATASET

σ_n	Index	LRTA	BM4D	KBR	FastHyDe	NLR-CPTD	SLRL4D	LTDL	WLRTR	GLF	LRTFD-FR	NG-Meet	**NLSSR**
20	PSNR↑	35.06	34.21	37.28	38.99	34.37	36.02	39.29	39.14	39.86	38.32	39.87	**40.14**
	SSIM↑	0.9464	0.9370	0.9720	0.9805	0.9474	0.9643	0.9807	0.9806	0.9835	0.9766	0.9839	**0.9845**
	FSIM↑	0.9696	0.9638	0.9836	0.9881	0.9706	0.9795	0.9888	0.9882	0.9898	0.9841	0.9903	**0.9906**
	ERGAS↓	72.76	79.74	55.87	47.93	78.17	66.48	45.06	45.92	42.61	50.74	41.54	**40.15**
	SAM↓	0.1220	0.1238	0.0709	0.0798	0.0887	0.0888	0.0696	0.0686	0.0674	0.0684	0.0659	**0.0627**
30	PSNR↑	32.71	31.85	34.81	37.19	33.57	35.06	37.06	36.85	37.72	36.90	37.72	**37.85**
	SSIM↑	0.9151	0.8957	0.9524	0.9709	0.9359	0.9546	0.9699	0.9690	0.9740	0.9679	0.9740	**0.9747**
	FSIM↑	0.9505	0.9402	0.9718	0.9823	0.9645	0.9737	0.9822	0.9812	0.9840	0.9793	**0.9846**	0.9844
	ERGAS↓	94.76	104.14	74.01	57.86	85.54	73.68	57.82	59.21	53.86	59.12	52.96	**52.10**
	SAM↓	0.1415	0.1497	0.0866	0.0907	0.1026	0.0967	0.0831	0.0836	0.0812	0.0790	0.0770	**0.0769**
40	PSNR↑	31.31	30.26	32.56	35.81	32.57	34.20	35.31	35.22	36.13	35.24	36.06	**36.38**
	SSIM↑	0.8883	0.8546	0.9206	0.9608	0.9186	0.9447	0.9573	0.9564	0.9639	0.9536	0.9629	**0.9659**
	FSIM↑	0.9354	0.9166	0.9549	0.9761	0.9557	0.9676	0.9746	0.9736	0.9779	0.9715	0.9784	**0.9794**
	ERGAS↓	111.13	124.64	96.04	67.17	95.81	80.78	70.35	71.10	64.34	71.00	63.99	**61.64**
	SAM↓	0.1530	0.1695	0.0996	0.1001	0.1183	0.1038	0.0944	0.0975	0.0937	0.0933	0.0897	**0.0877**
50	PSNR↑	30.24	29.09	31.62	34.71	31.51	33.45	34.18	33.90	34.95	33.75	34.83	**35.21**
	SSIM↑	0.8628	0.8151	0.9023	0.9506	0.8970	0.9344	0.9449	0.9423	0.9535	0.9358	0.9513	**0.9560**
	FSIM↑	0.9215	0.8938	0.9435	0.9699	0.9447	0.9614	0.9683	0.9654	0.9718	0.9622	0.9721	**0.9737**
	ERGAS↓	125.72	142.42	107.02	75.86	108.07	87.68	79.96	82.50	73.54	83.99	73.62	**70.52**
	SAM↓	0.1629	0.1854	0.1106	0.1084	0.1339	0.1100	0.1031	0.1108	0.1034	0.1090	0.0979	**0.0964**
75	PSNR↑	28.11	27.10	29.38	32.67	29.35	31.92	31.85	31.10	32.77	31.04	32.65	**32.95**
	SSIM↑	0.7935	0.7274	0.8405	0.9251	0.8415	0.9077	0.9094	0.8980	0.9264	0.8859	0.9235	**0.9291**
	FSIM↑	0.8884	0.8437	0.9058	0.9549	0.9158	0.9456	0.9493	0.9411	0.9563	0.9341	0.9565	**0.9580**
	ERGAS↓	160.64	178.69	138.39	95.28	138.31	103.89	104.24	113.36	94.07	114.45	94.49	**91.24**
	SAM↓	0.1816	0.2153	0.1348	0.1258	0.1656	0.1228	0.1196	0.1504	0.1224	0.1319	0.1183	**0.1146**
100	PSNR↑	26.83	25.80	28.16	31.21	28.27	30.71	30.20	29.11	31.17	28.87	31.20	**31.46**
	SSIM↑	0.7387	0.6550	0.7954	0.8997	0.8040	0.8788	0.8708	0.8465	0.8980	0.8277	0.8968	**0.9040**
	FSIM↑	0.8649	0.8041	0.8773	0.9409	0.8939	0.9290	0.9288	0.9159	0.9405	0.9001	0.9424	**0.9442**
	ERGAS↓	186.27	207.26	159.15	112.46	156.66	119.17	125.91	142.23	112.74	147.27	111.60	**108.21**
	SAM↓	0.1952	0.2372	0.1520	0.1411	0.1748	0.1326	0.1321	0.1826	0.1378	0.1520	0.1294	**0.1289**
Average	PSNR↑	30.71	29.72	32.30	35.10	31.60	33.56	34.65	34.22	35.43	34.02	35.39	**35.67**
	SSIM↑	0.8575	0.8141	0.8972	0.9480	0.8907	0.9308	0.9389	0.9321	0.9499	0.9246	0.9487	**0.9524**
	FSIM↑	0.9217	0.8937	0.9395	0.9687	0.9409	0.9595	0.9653	0.9609	0.9701	0.9552	0.9707	**0.9717**
	ERGAS↓	125.21	139.48	105.08	76.09	110.43	88.61	80.56	85.72	73.53	87.76	73.03	**70.64**
	SAM↓	0.1594	0.1802	0.1091	0.1077	0.1307	0.1091	0.1003	0.1156	0.1010	0.1056	0.0964	**0.0945**



(a) (b) (c) (d) (e) (f)

Fig. 7. PSNR values of each band on the PAC dataset at different noise levels. (a)–(f) $\sigma_n = 20$, 30, 40, 50, 75, and 100, respectively.



(a) (b) (c) (d) (e) (f) (g)

(h) (i) (j) (k) (l) (m) (n)

Fig. 8. Denoising results of the *PAC* image by different methods ($\sigma_n = 100$). (a) Original image (bands: 12, 42, and 63). (b) Noisy image. (c) LRTA [9]. (d) BM4D [29]. (e) KBR [27]. (f) FastHyDe [35]. (g) NLR-CPTD [58]. (h) SLRL4D [36]. (i) LTDL [5]. (j) WLRTR [6]. (k) GLF [32]. (l) LRTFD-FR [59]. (m) NG-Meet [7]. (n) **NLSSR**.

still suffer from undesirable ringing artifacts, while KBR and LTDL methods are prone to produce the oversmooth effect. While the NG-Meet method can effectively reduce noise, it tends to generate some blurring and can result in the loss of image details. In contrast, the proposed NLSSR algorithm not only achieves significant noise reduction, but also effectively suppresses undesirable ringing artifacts while preserving fine image structures and details.

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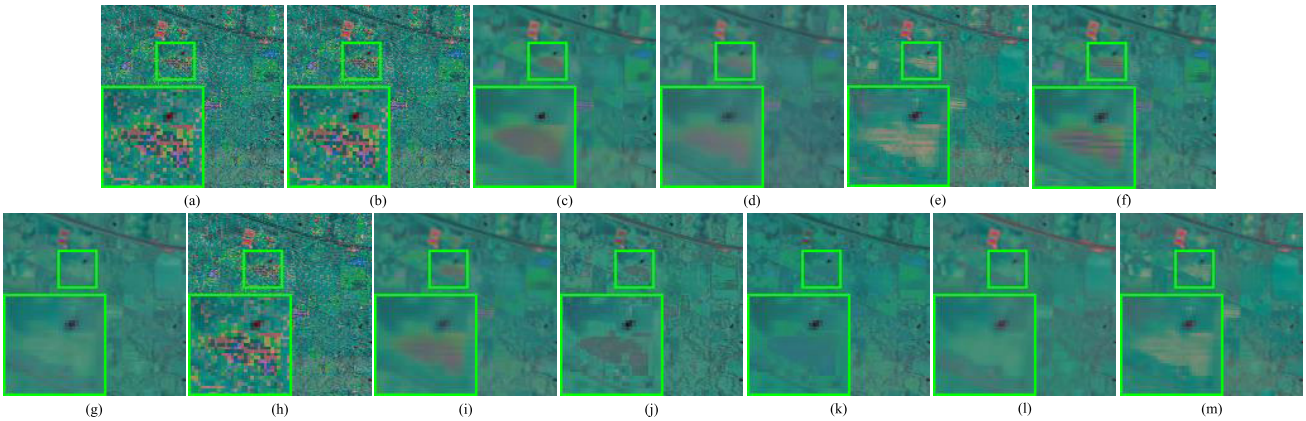


Fig. 9. Real denoising results of the *AVIRIS Indian Pines dataset*. (a) Noisy image (bands: 1, 83, and 91). (b) LRTA [9]. (c) BM4D [29]. (d) KBR [27]. (e) FastHyDe [35]. (f) NLR-CPTD [58]. (g) SLRL4D [36]. (h) LTDL [5]. (i) WLRTR [6]. (j) GLF [32]. (k) LRTFD-FR [59]. (l) NG-Meet [7]. (m) **NLSSR**.

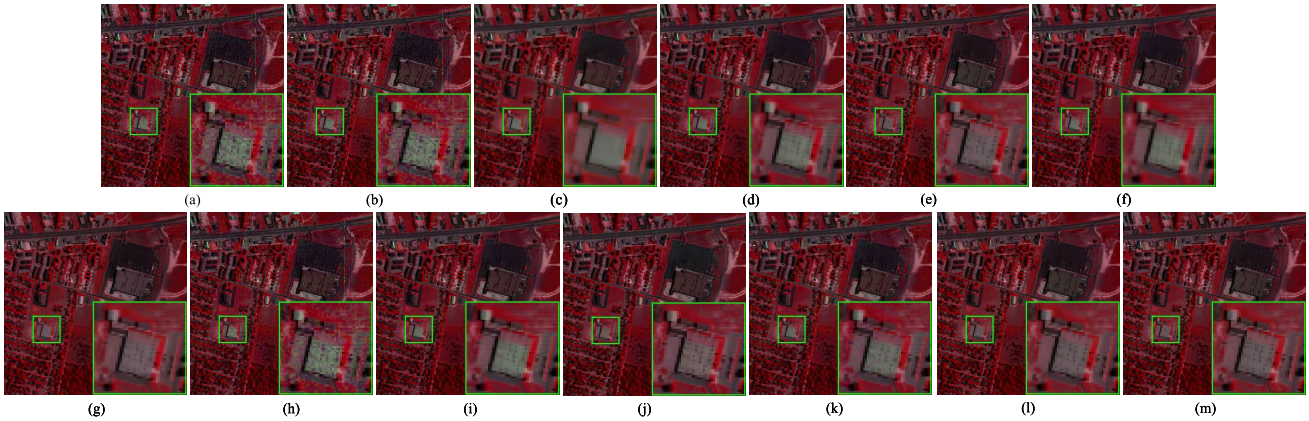


Fig. 10. Real denoising results of the *HYDICE Urban dataset*. (a) Noisy image (bands: 100, 196, and 206). (b) LRTA [9]. (c) BM4D [29]. (d) KBR [27]. (e) FastHyDe [35]. (f) NLR-CPTD [58]. (g) SLRL4D [36]. (h) LTDL [5]. (i) WLRTR [6]. (j) GLF [32]. (k) LRTFD-FR [59]. (l) NG-Meet [7]. (m) **NLSSR**.

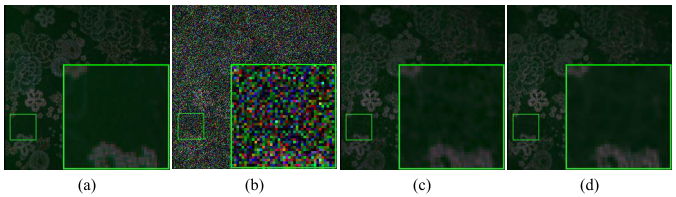
2) Simulated Experiments for PAC and WDC Datasets: PAC and WDC are two well-known HSI datasets for HSI denoising, which include 102 and 192 spectral bands, respectively. For simplicity, a subimage of size $366 \times 358 \times 102$ is cropped from the PAC dataset and a subimage of size $256 \times 256 \times 191$ is cropped from the WDC dataset. We compare the proposed NLSSR with 11 popular or state-of-the-art HSI denoising approaches, including LRTA [9], BM4D [29], KBR [27], FastHyDe [35], NLR-CPTD [58], SLRL4D [36], LTDL [5], WLRTR [6], GLF [32], LRTFD-FR [59], and NG-Meet [7]. These competing methods are both used to deal with high-band datasets (e.g., PAC dataset and WDC dataset) in their papers. Similar to testing the CAVE dataset, we exploit the default parameter setting or suggested parameters derived from their papers to obtain the denoising results. It is worth noting that LRTA and LRTFD-FR are the global spectral LR HSI denoising approaches, while BM4D, KBR, NLR-CPTD, LTDL, and WLTRTR are spatial nonlocal-based approaches. FastHyDe, NG-Meet, SLRL4D, and GLF are jointly using global spectral LR and NSS priors.

Five quantitative comparison results of the PAC dataset and the WDC dataset are shown in Tables II and III, respectively. It can be observed that the proposed NLSSR achieves better quantitative results than all compared methods (the only exception is when $\boldsymbol{\sigma}_n = 30$ for which NG-Meet is slightly higher FSIM than the proposed method on testing the WDC dataset). In particular, when HSIs suffer from heavy noise levels, the superiority of the proposed NLSSR over other compared methods makes greater through a large marginal. Fig. 7 shows the PSNR values of each band for the PAC dataset. The results show that our proposed NLSSR method (red line) achieves much higher PSNR values than the other compared methods. Additionally, Fig. 8 illustrates the visual comparison results of the PAC dataset. Our findings indicate that the LRTA and LRTFD-FR methods fail to produce satisfactory denoising results. While KBR, NLR-CPTD, SLRL4D, LTDL, and WLRTR methods can effectively remove noise, they struggle to reconstruct the spectral structure of HSIs. Conversely, BM4D, FastHyDe, and NG-Meet methods can effectively reconstruct the spatial and spectral information, but they tend to oversmooth images. In contrast, our proposed NLSSR method not only removes noise effectively, but also preserves both spatial and spectral information accurately.

B. Real Data Experiments

Before that, we demonstrated the effectiveness of the proposed NLSSR model for the simulated HSI denoising. In this section, we test two well-known real noisy HSI datasets to further verify the effectiveness of the proposed NLSSR

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(a) (b) (c) (d)

Fig. 11. Visual comparison results of the MSI *Cloth* by the BM and GBM methods ($\sigma_n = 75$). (a) Original image (bands: 1, 4, and 11). (b) Noisy image. (c) BM. (d) GBM.

algorithm. The first is the AVIRIS Indian Pines dataset[6] and the original size is $145 \times 145 \times 220$. Since 20 water absorption bands (104–108, 150-163, and 220 bands) do not have useful information, we discard them. Then the size of the AVIRIS Indian Pines dataset becomes $145 \times 145 \times 200$. The second is the HYDICE Urban dataset,[7] which includes 307×307 spatial pixels and 210 spectral bands. It is worth noting that noise degradations of these two datasets are not pure Gaussian white noise. The parameters of all competing methods are set the same as the simulated experiments.

The visual comparison results of the AVIRIS Indian Pines dataset and the HYDICE Urban dataset for different methods are shown in Figs. 9 and 10, respectively. For the AVIRIS Indian Pines dataset in Fig. 9, we can see that noises obviously exist in LRTA [9] and LTDL [5] methods. BM4D [29], FastHyDe [35], NLR-CPTD [58], and GLF [32] methods can effectively remove noise; however, they often exhibit undesirable visual artifacts. On the other hand, KBR [27], SLRL4D [36], WLRTR [6], LRTFD-FR [59], and NG-Meet [7] methods tend to oversmooth images and result in the loss of some image details. In contrast, our proposed NLSSR method not only eliminates noise effectively, but also preserves fine image details. As can be seen in Fig. 10 for the HYDICE Urban dataset, LRTA [9] and LTDL [5] methods cannot eliminate the noise effectively. BM4D [29], KBR [27], WLRTR [6], and NG-Meet [7] can completely remove the noise, but they usually oversmooth images. Although NLR-CPTD [58] and LRTFD-FR [59] methods achieve better visual results than the above methods, they still generate some undesirable ringing artifacts. FastHyDe [35], SLRL4D [36], GLF [32], and the proposed NLSSR achieve the best visual results. Moreover, we apply a well-known no-reference image quality metric, that is, BRISQUE [60], to evaluate the denoising performance of all competing approaches. It is worth noting that the smaller the BRISQUE value is, the better the denoising performance is. As illustrated in Figs. 9 and 10, the proposed NLSSR outperforms other competing methods by achieving the lowest BRISQUE values. To summarize, although the NLSSR model was initially developed with the assumption of Gaussian noise, it has proven to be advantageous for processing real noisy images.

[6]https://engineering.purdue.edu/~biehl/MultiSpec/

[7]https://www.erdc.usace.army.mil/Media/Fact-Sheets/Fact-Sheet-Article-View/Article/610433/hypercube/Scenes

TABLE IV

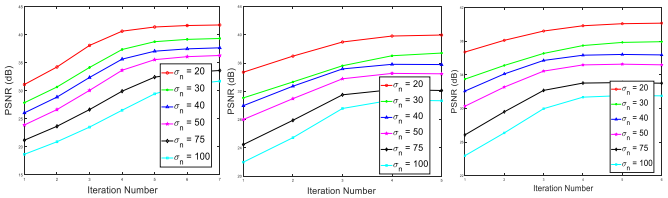
FIVE QUANTITATIVE RESULTS ARE OBTAINED BY HSI-SDECNN [61], MAC-NET [62], AND THE PROPOSED NLSSR METHODS ON THE WDC DATASET

σ_n	Methods	PSNR↑	SSIM↑	FSIM↑	ERGAS↓	SAM↓
10	HSI-SDeCNN [61]	40.04	0.9909	0.9838	41.89	0.0776
	MAC-Net [62]	42.40	0.9943	0.9909	32.74	0.0550
	NLSSR	**43.83**	**0.9957**	**0.9928**	**26.47**	**0.0455**
30	HSI-SDeCNN [61]	34.46	0.9701	0.9463	78.35	0.1216
	MAC-Net [62]	36.18	0.9792	0.9663	64.91	0.0985
	NLSSR	**37.85**	**0.9844**	**0.9747**	**52.10**	**0.0769**
50	HSI-SDeCNN [61]	31.76	0.9470	0.9045	106.32	0.1506
	MAC-Net [62]	33.52	0.9658	0.9437	88.12	0.1250
	NLSSR	**35.21**	**0.9737**	**0.9560**	**70.52**	**0.0964**
75	HSI-SDeCNN [61]	29.69	0.9183	0.8518	134.20	0.1768
	MAC-Net [62]	31.34	0.9433	0.9033	111.54	0.1453
	NLSSR	**32.95**	**0.9580**	**0.9291**	**91.24**	**0.1146**

TABLE V

PSNR/ERGAS COMPARISONS OF BM AND GBM METHODS FOR HSI DENOISING BY TESTING MSI *Cloth* FROM THE CAVE DATASET, PAC DATASET, AND WDC DATASET

	MSI *Cloth* of the CAVE Dataset					
Methods	20	30	40	50	75	100
BM	37.16/65.16	34.76/86.25	33.26/103.03	31.50/127.00	29.87/154.33	28.39/183.67
GBM	**37.39/63.56**	**35.08/83.25**	**33.42/101.15**	**31.78/122.92**	**30.09/149.62**	**28.51/181.02**
	PAC Dataset					
Methods	20	30	40	50	75	100
BM	39.77/44.17	37.05/60.26	35.62/70.95	34.29/82.67	32.13/106.07	30.67/125.41
GBM	**39.96/43.23**	**37.41/57.82**	**35.82/69.38**	**34.53/80.41**	**32.24/104.76**	**30.77/124.02**
	WDC Dataset					
Methods	20	30	40	50	75	100
BM	39.72/42.11	37.61/53.55	36.27/62.43	35.08/71.56	32.91/91.73	31.36/109.53
GBM	**40.14/40.16**	**37.85/52.10**	**36.42/61.41**	**35.26/70.08**	**33.07/89.99**	**31.53/107.43**



(a) (b) (c)

Fig. 12. Convergence behavior of the proposed NLSSR algorithm. (a) PSNR results versus iteration numbers for HSI denoising on the MSI *Toy* of the CAVE dataset. (b) PSNR results versus iteration numbers for HSI denoising on the PAC dataset. (c) PSNR results versus iteration numbers for HSI denoising on the WDC dataset.

C. Comparison With Deep-Learning Approach

Recent studies have indicated that deep-learning methods have great potential in HSI denoising. In this section, we evaluate our proposed NLSSR method against two state-of-the-art deep-learning approaches, namely HSI-SDeCNN [61] and MAC-Net [62]. Table IV presents the quantitative results of the PAC dataset obtained using HSI-SDeCNN, MAC-Net, and our proposed NLSSR methods. It is evident that our proposed NLSSR method achieves better quantitative results compared to the other two competing methods.

D. Effectiveness of the Proposed GBM

To demonstrate the effectiveness of the proposed GBM, we compare it with the existing BM method [17], [38]. The MSI *Cloth* from the CAVE dataset, the PAC dataset, and the WDC dataset are chosen as the experimental objects. We show the PSNR and ERGAS comparisons of BM and GBM methods for different noise levels are shown in Table V. It can be

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TABLE VI
AVERAGE RUNNING TIME (S) OF BM4D [29], KBR [27], NLR-CPTD [58], NG-MEET [7], SLRL4D [36], LTDL [5], WLRTR [6], GLF [32], AND THE PROPOSED NLSSR METHODS ON THE WDC AND PAC DATASETS

WDC Dataset									
Methods	**BM4D**	**KBR**	**NLR-CPTD**	**SLRL4D**	**LTDL**	**WLRTR**	**GLF**	**NG-Meet**	**NLSSR**
Time	618.71	3542.89	8386.18	393.16	15035.04	19770.22	582.47	88.73	**75.96**
PAC Dataset									
Methods	**BM4D**	**KBR**	**NLR-CPTD**	**NG-Meet**	**SLRL4D**	**LTDL**	**WLRTR**	**GLF**	**NLSSR**
Time	619.62	3583.26	7064.25	767.94	6393.05	16366.25	1239.37	**160.43**	192.1

seen that the proposed GBM achieves the best quantitative values (larger PSNR and smaller ERGAS) than the existing BM method. The visual comparison of BM and GBM methods is shown in Fig. 11, from which we can see that the BM method obviously produces undesirable visual artifacts with weaker quantitative values (lower PSNR and higher ERGAS values) obtained by 1, 4, and 11 bands (magnify the green windows). In contrast, the proposed GBM algorithm alleviates visual artifacts as much as possible. Therefore, this experiment demonstrates the effectiveness of the proposed GBM algorithm.

E. Convergence Analysis

It is challenging to give the theoretical convergence proof of the proposed NLSSR denoising algorithm because of the existence of the GBM operation and the nonconvex L_p minimization. Therefore, we have only provided empirical evidence to depict the convergence behavior of the proposed NLSSR denoising algorithm. Fig. 12 plots the evaluation of PSNR values versus iteration numbers for MSI *Toy* of the CAVE dataset, PAC dataset, and the WDC dataset with different noise levels. It can be seen that in both cases, with the increasing of iteration numbers, the PSNR curves monotonically increase and finally become flat and stable. Hence, the proposed NLSSR algorithm enjoys a good convergence.

F. Algorithm Complexity and Computational Time

The primary sources of computational complexity in the proposed NLSSR algorithm arise from solving the $\boldsymbol{B}$ and $\mathcal{Z}$ subproblems. The complexities associated with the $\boldsymbol{B}$ and $\mathcal{Z}$ subproblems are $O(HWC^2)$ and $O(nT_b+nmc^2)$, respectively. Here, T_b denotes the graph-based domain distance scheme utilized for identifying similar full-band patches. Hence, the total complexity of the proposed NLSSR algorithm for HSI denoising is $O(HWC^2+nT_b+nmc^2)$.

Moreover, we report the running time of the proposed NLSSR-based HSI denoising algorithm with many NSS-based HSI denoising methods. All experiments are conducted on an Intel[8] Core[9] i9-10920X with 64 GB memory and 3.50 Hz CPU PC under the MATLAB 2019b environment. We compare the proposed NLSSR with eight NSS-based HSI denoising approaches, including BM4D [29], KBR [27], NLR-CPTD [58], NG-Meet [7], SLRL4D [36], LTDL [5], WLRTR [6], and GLF [32]. We apply these NSS-based methods to test two HSI datasets (i.e., WDC and PAC), and the computational time of these NSS-based HSI denoising methods is shown in Table VI. It can be observed that by testing the WDC dataset, the proposed NLSSR uses less computational time than all competing methods. By testing the PAC dataset, the proposed NLSSR uses slightly more time compared to the NG-Meet method, while using far less time than other methods.

[8]Registered trademark.
[9]Trademarked.

VI. CONCLUSION

In this article, we have presented a new HSI denoising approach, named NLSSR, by jointly exploiting the global spectral and nonlocal structured sparsity priors of the HSI. We have proposed a graph-based domain distance strategy to compute the patch similarity of the HSI to alleviate the noise sensitivity of the BM operation. We have proposed an adaptive unidirectional LR dictionary with low complexity, which fully takes the difference of the intrinsic structure correlation among different modes of the nonlocal full-band tensor into consideration. Moreover, an effective algorithm has been proposed, which applies an alternating minimization strategy to solve the proposed NLSSR-based HSI denoising problem. The proposed NLSSR approach has been extensively tested by simulated and real data, and the results demonstrate that the proposed NLSSR outperforms many popular or state-of-the-art HSI denoising algorithms both quantitatively and qualitatively.

APPENDIX
PROOF OF THE THEOREM 1

Proof: The adaptive dictionary $\boldsymbol{D}_{\mathbf{G}_i}$ is constructed by (12). From the unitary property of $\boldsymbol{U}_{\mathbf{G}_i}$ and $\boldsymbol{V}_{\mathbf{G}_i}$, we have

$$\begin{aligned}
\left\|\bar{\boldsymbol{Z}}_i^{(2)}-\bar{\boldsymbol{X}}_i^{(2)}\right\|_F^2 &= \left\|\boldsymbol{D}_{\mathbf{G}_i}(\boldsymbol{R}_{\mathbf{G}_i}-\boldsymbol{A}_{\mathbf{G}_i})\right\|_F^2 \\
&= \left\|\boldsymbol{U}_{\mathbf{G}_i}\mathrm{diag}(\boldsymbol{R}_{\mathbf{G}_i}-\boldsymbol{A}_{\mathbf{G}_i})\boldsymbol{V}_{\mathbf{G}_i}^T\right\|_F^2 \\
&= \mathrm{Tr}\left(\boldsymbol{U}_{\mathbf{G}_i}\mathrm{diag}(\boldsymbol{R}_{\mathbf{G}_i}-\boldsymbol{A}_{\mathbf{G}_i})\boldsymbol{V}_{\mathbf{G}_i}^T\boldsymbol{V}_{\mathbf{G}_i}\mathrm{diag}(\boldsymbol{R}_{\mathbf{G}_i}-\boldsymbol{A}_{\mathbf{G}_i})\boldsymbol{U}_{\mathbf{G}_i}^T\right) \\
&= \mathrm{Tr}\left(\boldsymbol{U}_{\mathbf{G}_i}\mathrm{diag}(\boldsymbol{R}_{\mathbf{G}_i}-\boldsymbol{A}_{\mathbf{G}_i})\mathrm{diag}(\boldsymbol{R}_{\mathbf{G}_i}-\boldsymbol{A}_{\mathbf{G}_i})\boldsymbol{U}_{\mathbf{G}_i}^T\right) \\
&= \mathrm{Tr}\left(\mathrm{diag}(\boldsymbol{R}_{\mathbf{G}_i}-\boldsymbol{A}_{\mathbf{G}_i})\boldsymbol{U}_{\mathbf{G}_i}^T\boldsymbol{U}_{\mathbf{G}_i}\mathrm{diag}(\boldsymbol{R}_{\mathbf{G}_i}-\boldsymbol{A}_{\mathbf{G}_i})\right) \\
&= \mathrm{Tr}(\mathrm{diag}(\boldsymbol{R}_{\mathbf{G}_i}-\boldsymbol{A}_{\mathbf{G}_i})\mathrm{diag}(\boldsymbol{R}_{\mathbf{G}_i}-\boldsymbol{A}_{\mathbf{G}_i})) \\
&= \left\|\boldsymbol{R}_{\mathbf{G}_i}-\boldsymbol{A}_{\mathbf{G}_i}\right\|_F^2 \qquad (26)
\end{aligned}$$

where $\bar{\boldsymbol{Z}}_i^{(2)}=\boldsymbol{D}_{\mathbf{G}_i}\boldsymbol{R}_{\mathbf{G}_i}$ and $\bar{\boldsymbol{X}}_i^{(2)}=\boldsymbol{D}_{\mathbf{G}_i}\boldsymbol{A}_{\mathbf{G}_i}$. □

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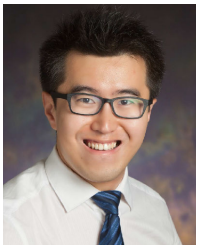
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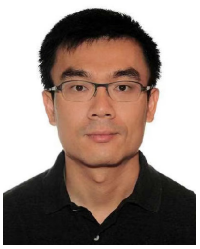
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